# Chapter 10 Part 3: Ksp

**1.** A $10.0\ g$ sample of $Pb(NO_3)_2(s)$ is dissolved in distilled water and an excess of $1.0\ M\ HCl(aq)$ is added. The $PbCl_2(s)$ formed is filtered and placed in an oven to dry. The final mass of the precipitate is significantly greater than predicted. Which of the following could account for the discrepancy?

(A) The sample of $Pb(NO_3)_2(s)$ was wet.

(B) The $K_{sp}$ of $PbCl_2(s)$ is extremely small.

(C) The precipitate was not dried to a constant mass. ✓

(D) A $3.0\ M\ HCl(aq)$ solution was used instead of a $1.0\ M\ HCl(aq)$ solution.

**2.** $BaCO_3(s) \rightleftarrows Ba^{2+}(aq) + CO_3{}^{2-}(aq) \qquad K_{sp} = 8.1 \times 10^{-9}$

The dissolution of $BaCO_3(s)$ in water is represented above. A $20.0\ mL$ sample of $1.0 \times 10^{-4}\ M\ Ba(NO_3)_2(aq)$ is mixed with an $80.0\ mL$ sample of $1.0 \times 10^{-4}\ M\ Na_2CO_3(aq)$. Which of the following correctly predicts and explains the outcome?

(A) A precipitate will form, because $Q = 1.0 \times 10^{-8}$.

(B) A precipitate will form, because $Q = 1.6 \times 10^{-9}$.

(C) A precipitate will not form, because $Q = 1.0 \times 10^{-8}$.

(D) A precipitate will not form, because $Q = 1.6 \times 10^{-9}$. ✓

**3.** $Ag^+(aq) + Cl^-(aq) \rightleftarrows AgCl(s)$

A student mixes dilute $AgNO_3(aq)$ with excess $NaCl(aq)$ to form $AgCl(s)$, as represented by the net ionic equation above. Which of the diagrams below best represents the ions that are present in significant concentrations in the solution? ($K_{sp}$ for AgCl is $1.8 \times 10^{-10}$.)

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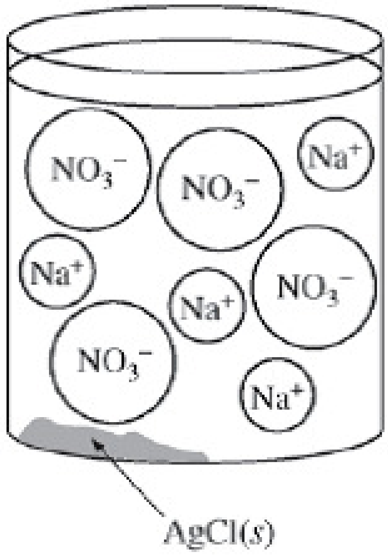
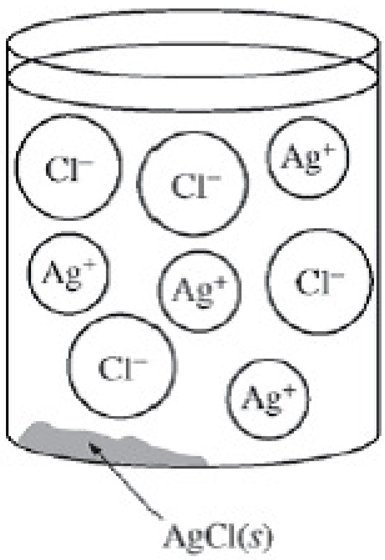
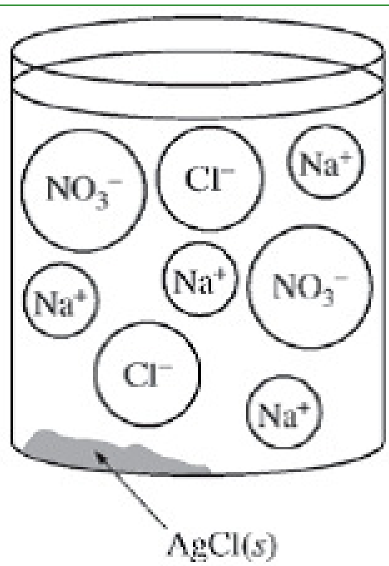
(A) 


(B) 


(C) 

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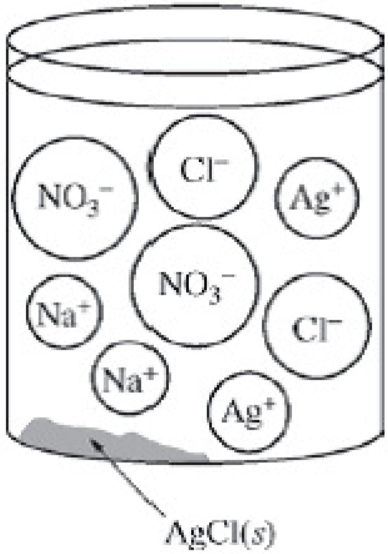
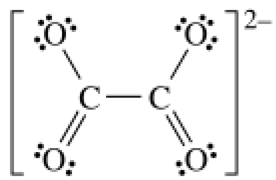
(D)



## Chapter 10 Part 3: Ksp

**4.** For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.



A Lewis electron-dot diagram of the oxalate ion, $C_2O_4{}^{2-}$, is shown.

(a) Identify the hybridization of the valence orbitals of either carbon atom in the oxalate ion.

(b) Silver oxalate, $Ag_2C_2O_4(s)$, is slightly soluble in water. The value of $K_{sp}$ for $Ag_2C_2O_4$ is $5.40 \times 10^{-12}$.

(i) Write the expression for the solubility-product constant, $K_{sp}$, for $Ag_2C_2O_4$.

(ii) Calculate the molar solubility of $Ag_2C_2O_4$ in neutral distilled water.

## Chapter 10 Part 3: Ksp

(iii) The molar solubility of $Ag_2C_2O_4$ increases when it is dissolved in $0.5 M$ $HClO_4(aq)$ instead of neutral distilled water. Write a balanced, net-ionic equation for the process that occurs between species in solution that contributes to the increased solubility of $Ag_2C_2O_4(aq)$ in $HClO_4(aq)$.

**Part A**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 ✓ |
|---|---|

The response provides the correct answer.

- $sp^2$

**Part B(i)**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 ✓ |
|---|---|

The response provides the correct answer.

- $K_{sp} = [Ag^+]^2[C_2O_4{}^{2-}]$

**Part B(ii)**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 ✓ |
|---|---|

The response provides the correct calculated value.

- $5.40 \times 10^{-12} = (2s)^2(s)$
  $5.40 \times 10^{-12} = 4s^3$
  $s = 1.11 \times 10^{-4}\ M$

## Chapter 10 Part 3: Ksp

### Part B(iii)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
| --- | --- |

The response provides a correct equation (state symbols not required).

- $C_2O_4{}^{2-}(aq) + H_3O^+(aq) \rightarrow HC_2O_4{}^-(aq) + H_2O(l)$

  OR

- $C_2O_4{}^{2-}(aq) + H^+(aq) \rightarrow HC_2O_4{}^-(aq)$

  OR

- $C_2O_4{}^{2-}(aq) + 2\ H_3O^+(aq) \rightarrow H_2C_2O_4(aq) + 2\ H_2O(l)a$

  OR

- $C_2O_4{}^{2-}(aq) + 2\ H^+(aq) \rightarrow H_2C_2O_4(aq)$

**5.** What is the molar solubility in water of $Ag_2CrO_4$ ? (The $K_{sp}$ for $Ag_2CrO_4$ is 8 x $10^{-12}$.)

(A) 8 x $10^{-12}$ $M$

(B) 2 x $10^{-12}$ $M$

(C) $\sqrt{4 \times 10^{-12}}\ M$

(D) $\sqrt[3]{4 \times 10^{-12}}\ M$

(E) $\sqrt[3]{2 \times 10^{-12}}\ M$

## Chapter 10 Part 3: Ksp

**6.** Several reactions are carried out using AgBr, a cream-colored silver salt for which the value of the solubility-product constant, $K_{sp}$ , is 5.0 x $10^{-13}$ at 298 K.

a. Write the expression for the solubility-product constant, $K_{sp}$ , of AgBr.

b. Calculate the value of $[Ag^+]$ in 50.0 mL of a saturated solution of AgBr at 298 K.

c. A 50.0 mL sample of distilled water is added to the solution described in part (b), which is in a beaker with some solid AgBr at the bottom. The solution is stirred and equilibrium is reestablished. Some solid AgBr remains in the beaker. Is the value of $[Ag^+]$ greater than, less than, or equal to the value you calculated in part (b) ? Justify your answer.

d. Calculate the minimum volume of distilled water, in liters, necessary to completely dissolve a 5.0 g sample of AgBr(*s*) at 298 K. (The molar mass of AgBr is 188 g $mol^{-1}$.)

e. A student mixes 10.0 mL of 1.5 x $10^{-4}$ *M* $AgNO_3$ with 2.0 mL of 5.0 x $10^{-4}$ *M* NaBr and stirs the resulting mixture. What will the student observe? Justify your answer with calculations.

f. The color of another salt of silver, AgI(s), is yellow. A student adds a solution of NaI to a test tube containing a small amount of solid, cream-colored AgBr. After stirring the contents of the test tube, the student observes that the solid in the test tube changes color from cream to yellow.

   i. Write the chemical equation for the reaction that occurred in the test tube.

   ii. Which salt has the greater value of $K_{sp}$ : AgBr or AgI ? Justify your answer.

**Part A**

1 point is earned for the correct expression (ion charges must be present; parentheses instead of square brackets not accepted).

*Ksp* = [Ag+][Br−]

| 0 | 1 ✓ |
|---|---|

The student response earns one of the following points:

1 point is earned for the correct expression (ion charges must be present; parentheses instead of square brackets not accepted).

*Ksp* = [Ag+][Br−]

**Part B**

1 point is earned for the correct value with support work (units not necessary).

## Chapter 10 Part 3: Ksp

Let *x* = equilibrium concentration of Ag+ (and of Br−).

Then *Ksp* = 5.0 x 10−13 = *x*2 ⇒ *x* = 7.1 x 10−7 *M*

| 0 | 1 |
|---|---|

The student response earns one of the following points:

1 point is earned for the correct value with support work (units not necessary).

Let *x* = equilibrium concentration of Ag+ (and of Br−).

Then *Ksp* = 5.0 x 10−13 = *x*2 ⇒ *x* = 7.1 x 10−7 *M*

**Part C**

1 point is earned for the correct answer with justification.

The value of [Ag+] after addition of distilled water is equal to the value in part (b). The concentration of ions in solution in equilibrium with a solid does not depend on the volume of the solution.

| 0 | 1 |
|---|---|

The student response earns one of the following points:

1 point is earned for the correct answer with justification.

The value of [Ag+] after addition of distilled water is equal to the value in part (b). The concentration of ions in solution in equilibrium with a solid does not depend on the volume of the solution.

**Part D**

1 point is earned for the calculation of moles of dissolved AgBr.

$$5.0 \text{ g AgBr} \times \frac{1 \text{ mol AgBr}}{188 \text{ g AgBr}} = 0.0266 \text{ mol AgBr}$$

1 point is earned for the correct answer for the volume of water.

$$\frac{0.0266 \text{ mol}}{\text{V}} = 7.1 \times 10^{-7} \text{mol L}^{-1} \Rightarrow \text{V} = 3.7 \times 10^{4} \text{ L}$$

| 0 | 1 | 2 |
|---|---|---|

## Chapter 10 Part 3: Ksp

The student response earns two of the following points:

1 point is earned for the calculation of moles of dissolved AgBr.

$$5.0 \text{ g AgBr} \times \frac{1 \text{ mol AgBr}}{188 \text{ g AgBr}} = 0.0266 \text{ mol AgBr}$$

1 point is earned for the correct answer for the volume of water.

$$\frac{0.0266 \text{ mol}}{\text{V}} = 7.1 \times 10^{-7} \text{mol L}^{-1} \Rightarrow \text{V} = 3.7 \times 10^{4} \text{ L}$$

**Part E**

1 point is earned for calculation of concentration of ions.

$$[\text{Ag}^{+}] = \frac{(10.0 \text{ mL})(1.5\times10^{-4} M)}{12.0 \text{ mL}} = 1.3 \times 10^{-4} M$$

$$[\text{Br}^{-}] = \frac{(2.0 \text{ mL})(5.0\times10^{-4} M)}{12.0 \text{ mL}} = 8.3 \times 10^{-5} M$$

1 point is earned for calculation of $Q$ and conclusion based on comparison between $Q$ and *Ksp*.

$Q$ = [Ag+][Br−] = (1.3 x 10-4 $M$)(8.3 x 10-5 $M$) = 1.1 x 10-8

1 point is earned for indicating the precipitation of AgBr.

1.1 x 10-8 > 5.0 x 10-13, ∴ a precipitate will form.

| 0 | 1 | 2 | 3 ✓ |
|---|---|---|---|

The student response earns three of the following points:

1 point is earned for calculation of concentration of ions.

$$[\text{Ag}^{+}] = \frac{(10.0 \text{ mL})(1.5\times10^{-4} M)}{12.0 \text{ mL}} = 1.3 \times 10^{-4} M$$

$$[\text{Br}^{-}] = \frac{(2.0 \text{ mL})(5.0\times10^{-4} M)}{12.0 \text{ mL}} = 8.3 \times 10^{-5} M$$

1 point is earned for calculation of $Q$ and conclusion based on comparison between $Q$ and *Ksp*.

$Q$ = [Ag+][Br−] = (1.3 x 10-4 $M$)(8.3 x 10-5 $M$) = 1.1 x 10-8

1 point is earned for indicating the precipitation of AgBr.

1.1 x 10-8 > 5.0 x 10-13, ∴ a precipitate will form.

**Part F**

- 1 point is earned for the correct equation.

## Chapter 10 Part 3: Ksp

$AgBr(s) + I^-(aq) \rightarrow AgI(s) + Br^-(aq)$

*OR*

$AgBr(s) + NaI(aq) \rightarrow AgI(s) + NaBr(aq)$

- 1 point is earned for the correct choice with justification.

  AgBr has the greater value of *Ksp*. The precipitate will consist of the less soluble salt when both $I^-(aq)$ and $Br^-(aq)$ are present. Because the color of the precipitate in the test tube turns yellow, it must be $AgI(s)$ that precipitates; therefore *Ksp* for AgBr must be greater than *Ksp* for AgI.

  *OR*

  *Keq* for the displacement reaction is $\frac{K_{sp} \text{ of AgBr}}{K_{sp} \text{ of AgI}}$ . Because yellow AgI forms, *Keq* > 1; therefore *Ksp* of AgBr > *Ksp* of AgI.

| 0 | 1 | 2 |
|---|---|---|

The student response earns two of the following points:

- 1 point is earned for the correct equation.

$AgBr(s) + I^-(aq) \rightarrow AgI(s) + Br^-(aq)$

*OR*

$AgBr(s) + NaI(aq) \rightarrow AgI(s) + NaBr(aq)$

- 1 point is earned for the correct choice with justification.

AgBr has the greater value of *Ksp*. The precipitate will consist of the less soluble salt when both $I^-(aq)$ and $Br^-(aq)$ are present. Because the color of the precipitate in the test tube turns yellow, it must be $AgI(s)$ that precipitates; therefore *Ksp* for AgBr must be greater than *Ksp* for AgI.

*OR*

*Keq* for the displacement reaction is $\frac{K_{sp} \text{ of AgBr}}{K_{sp} \text{ of AgI}}$ . Because yellow AgI forms, *Keq* > 1; therefore *Ksp* of AgBr > *Ksp* of AgI.

## Chapter 10 Part 3: Ksp

**7.** **Directions**: **For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.**

A student investigates various properties of silver compounds in the lab. First, the student carries out the following procedure to determine the concentration of $Ag^+(aq)$ in a solution of $AgNO_3(aq)$. The student adds an excess of $8.0\ M\ HCl(aq)$ to $200.0\ mL$ of the solution, causing $AgCl(s)$ to precipitate. The precipitate is separated from the solution by filtering it through a pre-weighed piece of filter paper. The precipitate and filter paper are rinsed with distilled water, dried, and weighed. The data from the experiment are given in the table below.

| | |
|---|---|
| Initial mass of filter paper | $0.851\ g$ |
| Mass of filter paper + $AgCl(s)$ precipitate after drying | $2.311\ g$ |

(a) Calculate the number of moles of $AgCl(s)$ that precipitated. (The molar mass of $AgCl$ is $143\ g/mol$).

(b) Assuming that all of the $Ag^+(aq)$ ions precipitated, calculate the original molarity of $Ag^+(aq)$ in the solution before $HCl(aq)$ was added.

(c) The calculated concentration of $Ag^+(aq)$ from the student's data is higher than the actual concentration of the original solution. The student claims that the calculated concentration is too high because the original solution contained some $Na^+(aq)$ impurities that resulted in some $NaCl(s)$ precipitating along with the $AgCl(s)$. Do you agree or disagree? Justify your answer.

After the experiment, the student learns that the precipitate, $AgCl(s)$, is slightly soluble in water. To determine the solubility, the student mixes a sample of the precipitate with warm, distilled water to make a saturated solution. The student finds the following equation for the dissolution of $AgCl(s)$ in water and a $K_{sp}$ value at the temperature of the water.

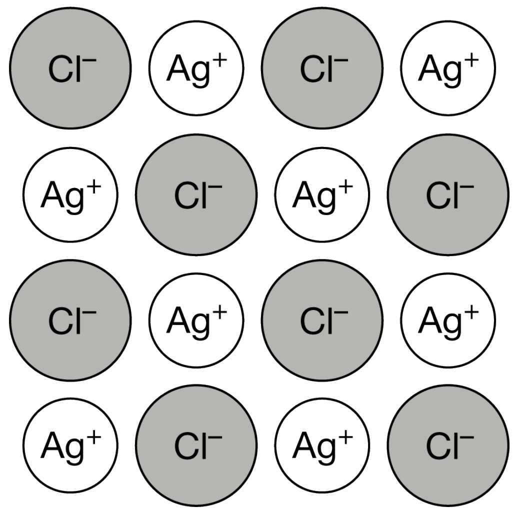
$$AgCl(s) \rightleftarrows Ag^+(aq) + Cl^-(aq) \qquad K_{sp} = 4.0 \times 10^{-10}$$

(d) Calculate $[Ag^+]$ in the student's solution.

(e) The value of $K_{sp}$ of another silver salt, $AgBr(s)$, is $5.0 \times 10^{-13}$. If separate, saturated solutions of $AgBr$ and $AgCl$ are prepared, which solution will have the greater $[Ag^+]$? Justify your answer.

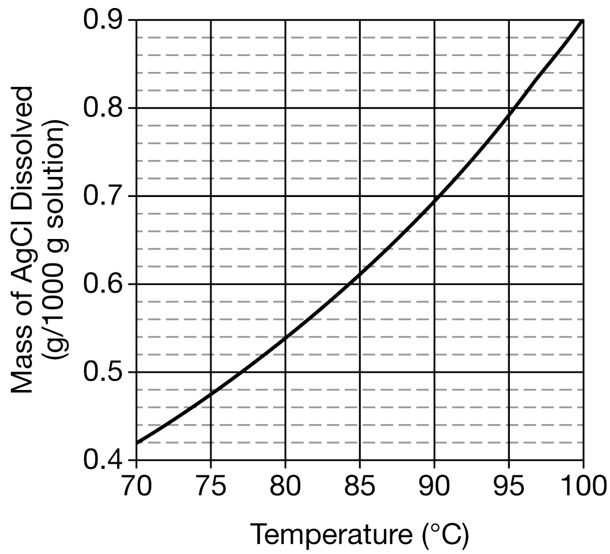
(f) The student finds the following particulate model of $AgCl(s)$. Assuming the crystal structures are similar, how should the student modify the model to represent $AgBr(s)$? Justify your answer in terms of the radius and arrangement of electrons in the ions.

## Chapter 10 Part 3: Ksp




(g) The enthalpy of dissolution of $\mathrm{AgCl}(s)$ in water is a positive value. Based on this information, the student claims that the energy released when separating the ions in $\mathrm{AgCl}(s)$is greater than the energy released during the hydration of $\mathrm{Ag}^{+}$ and $\mathrm{Cl}^{-}$ by water molecules. Do you agree or disagree with the student's claim? Justify your answer.

The student then conducts an experiment to determine the solubility of $\mathrm{AgCl}$ in a $2.5\ M\ \mathrm{NaCl}$ solution at various temperatures. Some $\mathrm{AgCl}(s)$ remains in the container at all temperatures studied. The data are plotted on the graph below.




(h) A $1000.$ g sample of a saturated solution at $95^{\circ}\mathrm{C}$ is cooled to $85^{\circ}\mathrm{C}$. Determine the mass of $\mathrm{AgCl}(s)$ that would precipitate out of the solution.

(i) The specific heat capacity of the solution is $3.3\ \mathrm{J}/(\mathrm{g}\cdot{}^{\circ}\mathrm{C})$ . How much heat is released when $1000.$ g of the solution is cooled from $95^{\circ}\mathrm{C}$ to $85^{\circ}\mathrm{C}$ ?

## Chapter 10 Part 3: Ksp

$$\begin{array}{lll} AgCl(s) \rightleftarrows Ag^+(aq) + Cl^-(aq) & & K_{sp} = 4.0 \times 10^{-10} \\ Ag^+(aq) + 2\,Cl^-(aq) \rightleftarrows [AgCl_2]^-(aq) & & K_2 = 1.8 \times 10^5 \\ \hline AgCl(s) + Cl^-(aq) \rightleftarrows [AgCl_2]^-(aq) & & K_3 = ? \end{array}$$

(j) The student finds that $\mathrm{AgCl}$ is more soluble in $\mathrm{NaCl}(aq)$ than in water because of the series of reactions above. Determine the value of $K_3$ for the overall reaction.

To further investigate the reactivity of silver, the student submerges a piece of metallic scandium, $\mathrm{Sc}(s)$, in a solution of $\mathrm{AgNO_3}(aq)$ and observes that the following reaction occurs.

$\mathrm{Sc}(s) + 3\,\mathrm{AgNO_3}(aq) \rightarrow 3\,\mathrm{Ag}(s) + \mathrm{Sc(NO_3)_3}(aq)$

(k) Write the balanced net-ionic equation for the reaction.

(l) The mass of the $\mathrm{Sc}(s)$ decreases significantly as it sits in the $\mathrm{AgNO_3}$ solution. The student claims that the mass decreases because scandium ions dissolve into the solution as the experiment progresses. Does the student's claim accurately explain the decrease in the mass of the $\mathrm{Sc}(s)$? Justify your answer.

**Part A**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 ✓ |
|---|---|

The response shows a correct calculation of moles.

- $2.311 \text{ g} - 0.851 \text{ g} = 1.460 \text{ g AgCl}(s)$

$1.460\,\mathrm{g\ AgCl}(s) \times \frac{1\,\mathrm{mol\ AgCl}(s)}{143\,\mathrm{g\ AgCl}(s)} = 1.02 \times 10^{-2}\,\mathrm{mol}$

**Part B**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 ✓ |
|---|---|

The response shows a correct calculation of molarity.

- $\frac{1.02\times10^{-2}\,\mathrm{mol}}{0.2000\,\mathrm{L}} = 5.1 \times 10^{-2}\,M$

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**Part C**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides the correct choice and a valid justification equivalent to the following.

· Disagree. $NaCl$ is soluble in water and would not have precipitated out of solution.

**Part D**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response shows a correct calculation of concentration.

· $4.0 \times 10^{-10} = [Ag^+][Cl^-] = x^2$

$[Ag^+] = x = \sqrt{4.0 \times 10^{-10}}\, M = 2.0 \times 10^{-5} M$

**Part E**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides both the correct answer and a valid justification equivalent to one of the following.

· The $AgCl$ solution will have the greater $[Ag^+]$. $AgCl$ has a larger value of $K_{sp}$ than $AgBr$, indicating a higher concentration of each ion in a saturated solution since both compounds have $1 : 1$ ion stoichiometry.

OR

· The $AgCl$ solution will have the greater $[Ag^+]$. For a saturated solution of $AgBr$, A $[Ag^+] = x = \sqrt{5.0 \times 10^{-13}}\, M = 7.1 \times 10^{-7} M$ , which is less than the $[Ag^+]$ for a saturated solution of $AgCl$.

**Part F**

## Chapter 10 Part 3: Ksp

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response meets both of the following criteria.

- The response indicates that the $Br^-$ ions should be larger.
- The response provides the explanation that $Br^-$ has a larger ionic radius than $Cl^-$ due to additional occupied electron shells ($n = 4$ versus $n = 3$ ).

**Part G**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides the correct choice and a valid justification equivalent to the following.

· Disagree. Overcoming attractions between ions requires the input of energy. Energy is absorbed, not released, when separating the ions in $AgCl$ .

**Part H**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides a correct calculation of mass.

· $0.79 \text{ g} - 0.61 \text{ g} = 0.18 \text{ g}$

**Part I**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

## Chapter 10 Part 3: Ksp

The response provides a correct calculation of the heat released.

$\cdot\, q = mc\Delta T = (1000.\,\mathrm{g})(3.3\,\frac{\mathrm{J}}{(\mathrm{g}\cdot{}^{\circ}\mathrm{C})})(85\,{}^{\circ}\mathrm{C} - 95\,{}^{\circ}\mathrm{C}) = -33,000\,\mathrm{J} = -33\,\mathrm{kJ}$

**Part J**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides the correct value.

$\cdot\, K_3 = \frac{[[\mathrm{AgCl_2}]^-]}{[\mathrm{Cl^-}]} = (K_{sp})(K_2) = (4.0 \times 10^{-10})(1.8 \times 10^{5}) = 7.2 \times 10^{-5}$

**Part K**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides the correctly balanced net-ionic equation.

$\cdot\, \mathrm{Sc}(s) + 3\,\mathrm{Ag}^{+}(aq) \rightarrow 3\,\mathrm{Ag}(s) + \mathrm{Sc}^{3+}(aq)$

(State symbols are not required.)

**Part L**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides both the correct answer and a valid justification equivalent to the following.

· Yes. As $\mathrm{Sc}(s)$ atoms are oxidized, they become ions that can be hydrated by water molecules. The mass of the solid metal decreases as these ions are formed and disperse in the solution.

## Chapter 10 Part 3: Ksp

**8. Directions: For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.**

A student investigates various properties of silver compounds in the lab. First, the student carries out the following procedure to determine the concentration of $Ag^{+}(aq)$ in a solution of $AgNO_3(aq)$. The student adds an excess of $2.0\ M\ HCl(aq)$ to $300.0\ mL$ of the solution, causing $AgCl(s)$ to precipitate. The precipitate is separated from the solution by filtering it through a pre-weighed piece of filter paper. The precipitate and filter paper are rinsed with distilled water, dried, and weighed. The data from the experiment are given in the table below.

| | |
|---|---|
| Initial mass of filter paper | $0.650\ g$ |
| Mass of filter paper + $AgCl(s)$ precipitate after drying | $0.793\ g$ |

(a) Calculate the number of moles of $AgCl(s)$ that precipitated. (The molar mass of $AgCl$ is $143\ g/mol$).

(b) Assuming that all of the $Ag^{+}(aq)$ ions precipitated, calculate the original molarity of $Ag^{+}(aq)$ in the solution before $HCl(aq)$ was added.

(c) The calculated concentration of $Ag^{+}(aq)$ from the student's data is lower than the actual concentration of the original solution. The student claims that the calculated concentration is too low because some of the $AgCl(s)$ passed through the filter paper. Do you agree or disagree? Justify your answer.

After the experiment, the student learns that the precipitate, $AgCl(s)$, is slightly soluble in water. To determine the solubility, the student mixes a sample of the precipitate with warm, distilled water to make a saturated solution. The student finds the following equation for the dissolution of $AgCl(s)$ in water and a $K_{sp}$ value at the temperature of the water.

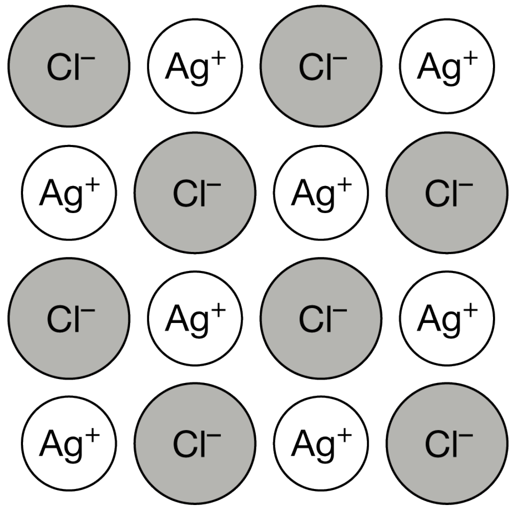
$$AgCl(s) \rightleftarrows Ag^{+}(aq) + Cl^{-}(aq) \qquad K_{sp} = 9.0 \times 10^{-10}$$

(d) Calculate $[Ag^{+}]$ in the student's solution.

(e) The value of $K_{sp}$ of another silver salt, $AgF(s)$, is $2.0 \times 10^{2}$. If separate, saturated solutions of $AgF(s)$ and $AgCl(s)$ are prepared, which solution will have the greater $[Ag^{+}]$? Justify your answer.

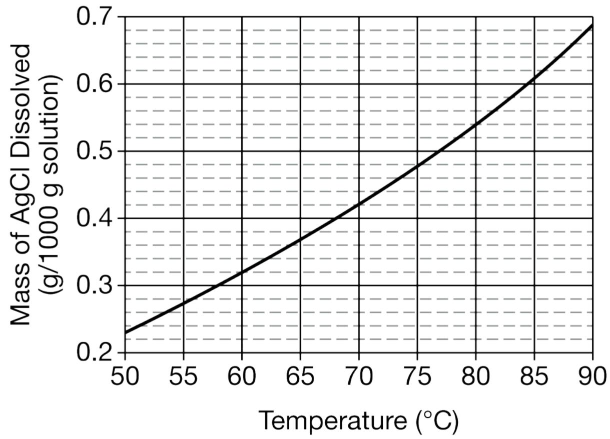
(f) The student finds the following particulate model of $AgCl(s)$. Assuming the crystal structures are similar, how should the student modify the model to represent $AgF(s)$? Justify your answer in terms of the radius and arrangement of electrons in the ions.

**Chapter 10 Part 3: Ksp**




(g) The enthalpy of dissolution of $AgCl(s)$ in water is a positive value. Based on this information, the student claims that the energy required to separate the ions in $AgCl(s)$ is equal to the energy released during the hydration of $Ag^{+}$ and $Cl^{-}$ by water molecules. Do you agree or disagree with the student's claim? Justify your answer.

The student then conducts an experiment to determine the solubility of $AgCl$ in a $2.5\ M$ $NaCl$ solution at various temperatures. Some $AgCl(s)$ remains in the container at all temperatures studied. The data are plotted on the graph below.




(h) A $1000.$ g sample of a saturated solution at $85^{\circ}C$ is cooled to $50.^{\circ}C$. Determine the mass of $AgCl(s)$ that would precipitate out of the solution.

(i) The specific heat capacity of the solution is $3.4\ J/(g \cdot {}^{\circ}C)$ . How much heat is released when $1000.$ g of the solution is cooled from $85^{\circ}C$ to $50.^{\circ}C$?

## Chapter 10 Part 3: Ksp

$$\begin{array}{lcll} AgCl(s) & \rightleftarrows & Ag^{+}(aq) + Cl^{-}(aq) & K_{sp} = 9.0 \times 10^{-10} \\ Ag^{+}(aq) + 2\,Cl^{-}(aq) & \rightleftarrows & [AgCl_2]^{-}(aq) & K_2 = 1.1 \times 10^{5} \\ \hline AgCl(s) + Cl^{-}(aq) & \rightleftarrows & [AgCl_2]^{-}(aq) & K_3 = ? \end{array}$$

(j) The student finds that $\mathrm{AgCl}$ is more soluble in $\mathrm{NaCl}(aq)$ than in water because of the series of reactions above. Determine the value of $K_3$ for the overall reaction.

To further investigate the reactivity of silver, the student submerges a piece of metallic tin, $\mathrm{Sn}(s)$, in a solution of $\mathrm{AgNO_3}(aq)$ and observes that the following reaction occurs.

$\mathrm{Sn}(s) + 2\,\mathrm{AgNO_3}(aq) \rightarrow 2\,\mathrm{Ag}(s) + \mathrm{Sn(NO_3)_2}(aq)$

(k) Write the balanced net-ionic equation for the reaction.

(l) The mass of the $\mathrm{Sn}(s)$ decreases significantly as it sits in the $\mathrm{AgNO_3}$ solution. The student claims that the mass decreases because the atoms in $\mathrm{Sn}(s)$ are converted to atoms of $\mathrm{Ag}(s)$ as the experiment progresses, and $\mathrm{Ag}(s)$ has a smaller molar mass than $\mathrm{Sn}(s)$ has. Does the student's claim accurately explain the decrease in the mass of the $\mathrm{Sn}(s)$? Justify your answer.

**Part A**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response shows a correct calculation of moles.

· $0.793\text{ g} - 0.650\text{ g} = 0.143\text{ g AgCl}(s) 0.143\,\text{g AgCl}(s) \times \frac{1\,\text{mol AgCl}(s)}{143\,\text{g AgCl}(s)} = 1.00 \times 10^{-3}\text{ mol}$

**Part B**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response shows a correct calculation of molarity.

· $\frac{1.00\times10^{-3}\,\text{mol}}{0.3000\,\text{L}} = 3.33 \times 10^{-3}\ M$

# Chapter 10 Part 3: Ksp

**Part C**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides the correct choice and a valid justification equivalent to the following.

· Agree. If some $\mathrm{AgCl}(s)$ was not captured by the filter paper, the student would not be weighing all of the precipitate. The calculated number of moles of $\mathrm{AgCl}(s)$, and therefore the calculated concentration of $\mathrm{Ag}^{+}(aq)$, would be too low.

**Part D**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response shows a correct calculation of concentration.

· $9.0 \times 10^{-10} = [\mathrm{Ag}^{+}][\mathrm{Cl}^{-}] = x^2$

$[\mathrm{Ag}^{+}] = x = \sqrt{9.0 \times 10^{-10}}\, M = 3.0 \times 10^{-5} M$

**Part E**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides both the correct answer and a valid justification equivalent to one of the following.

· The $\mathrm{AgF}$ solution will have the greater $[\mathrm{Ag}^{+}]$. $\mathrm{AgF}$ has a larger value of $K_{sp}$ than $\mathrm{AgCl}$, indicating a higher concentration of each ion in a saturated solution since both compounds have $1 : 1$ ion stoichiometry.

OR

· The $\mathrm{AgF}$ solution will have the greater $[\mathrm{Ag}^{+}]$. For a saturated solution of $\mathrm{AgF}$, $[\mathrm{Ag}^{+}] = x = \sqrt{2.0 \times 10^{2}}\, M = 1.4 \times 10^{1} M$, which is greater than the $[\mathrm{Ag}^{+}]$ for a saturated solution of $\mathrm{AgCl}$.

**Part F**

## Chapter 10 Part 3: Ksp

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
| --- | --- |

The response meets both of the following criteria.

- The response indicates that the $F^{-}$ ions should be smaller.
- The response provides the explanation that $F^{-}$ has a smaller ionic radius than $Cl^{-}$ due to fewer occupied electron shells ($n = 2$ versus $n = 3$).

### Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
| --- | --- |

The response provides the correct choice and a valid justification equivalent to the following.

· Disagree. Since the enthalpy of solution is positive, more energy is absorbed by the system to overcome ion-ion attractions than is released when the ions were hydrated by water molecules. Therefore, the student's claim is incorrect as it would result in an enthalpy of solution equal to $0\ \mathrm{kJ/mol}_{rxn}$.

### Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
| --- | --- |

The response provides a correct calculation of mass.

· $0.61\ \mathrm{g} - 0.23\ \mathrm{g} = 0.38\ \mathrm{g}$

### Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
| --- | --- |

## Chapter 10 Part 3: Ksp

The response provides a correct calculation of the heat released.

· $q = mc\Delta T = (1000.\,\mathrm{g})(3.4\,\frac{\mathrm{J}}{(\mathrm{g}\cdot{}^{\circ}\mathrm{C})})(50\,^{\circ}\mathrm{C} - 85\,^{\circ}\mathrm{C}) = -120,000\,\mathrm{J} = -120\,\mathrm{kJ}$

**Part J**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
| --- | --- |

The response provides the correct value.

· $K_3 = \frac{[[\mathrm{AgCl_2}]^-]}{[\mathrm{Cl}^-]} = (K_{sp})(K_2) = (9.0 \times 10^{-10})(1.1 \times 10^5) = 9.9 \times 10^{-5}$

**Part K**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
| --- | --- |

The response provides the correctly balanced net-ionic equation.

· $\mathrm{Sn}(s) + 2\,\mathrm{Ag}^+(aq) \rightarrow 2\,\mathrm{Ag}(s) + \mathrm{Sn}^{2+}(aq)$ (State symbols are not required.)

**Part L**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
| --- | --- |

The response provides both the correct answer and a valid justification equivalent to the following.

· No. This redox reaction involves the transfer of electrons between species, but none of the species change their elemental identity.

## Chapter 10 Part 3: Ksp

**9.** **Directions: For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.**

A student investigates various properties of silver compounds in the lab. First, the student carries out the following procedure to determine the concentration of $Ag^{+}(aq)$ in a solution of $AgNO_3(aq)$. The student adds an excess of $3.0\ M\ HCl(aq)$ to $100.0\ mL$ of the solution, causing $AgCl(s)$ to precipitate. The precipitate is separated from the solution by filtering it through a pre-weighed piece of filter paper. The precipitate and filter paper are rinsed with distilled water, dried, and weighed. The data from the experiment are given in the table below.

| | |
|---|---|
| Initial mass of filter paper | $0.718\ g$ |
| Mass of filter paper $+AgCl(s)$ precipitate after drying | $3.578\ g$ |

(a) Calculate the number of moles of $AgCl(s)$ that precipitated. (The molar mass of $AgCl$ is $143\ g/mol$).

(b) Assuming that all of the $Ag^{+}(aq)$ ions precipitated, calculate the original molarity of $Ag^{+}(aq)$ in the solution before $HCl(aq)$ was added.

(c) The calculated concentration of $Ag^{+}(aq)$ from the student's data is lower than the actual concentration of the original solution. The student claims that the calculated concentration is too low because the filter paper and $AgCl(s)$ precipitate were not fully dry. Do you agree or disagree? Justify your answer.

After the experiment, the student learns that the precipitate, $AgCl(s)$, is slightly soluble in water. To determine the solubility, the student mixes a sample of the precipitate with cold, distilled water to make a saturated solution. The student finds the following equation for the dissolution of $AgCl(s)$ in water and a $K_{sp}$ value at the temperature of the water.

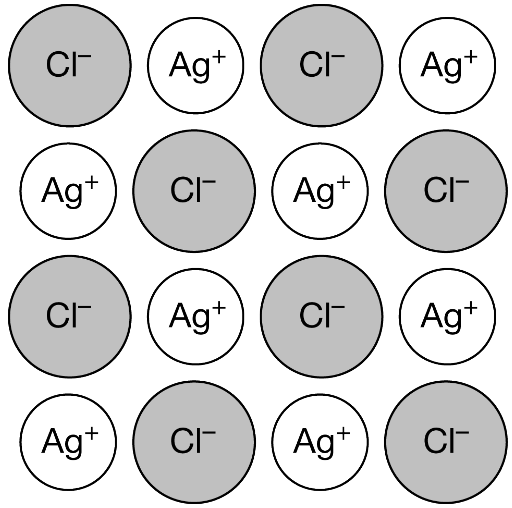
$$AgCl(s) \rightleftarrows Ag^{+}(aq) + Cl^{-}(aq) \quad K_{sp} = 1.0 \times 10^{-10}$$

(d) Calculate $[Ag^{+}]$ in the student's solution.

(e) The value of $K_{sp}$ of another silver salt, $AgI(s)$, is $8.3 \times 10^{-17}$. If separate, saturated solutions of $AgI$ and $AgCl$ are prepared, which solution will have the greater $[Ag^{+}]$? Justify your answer.

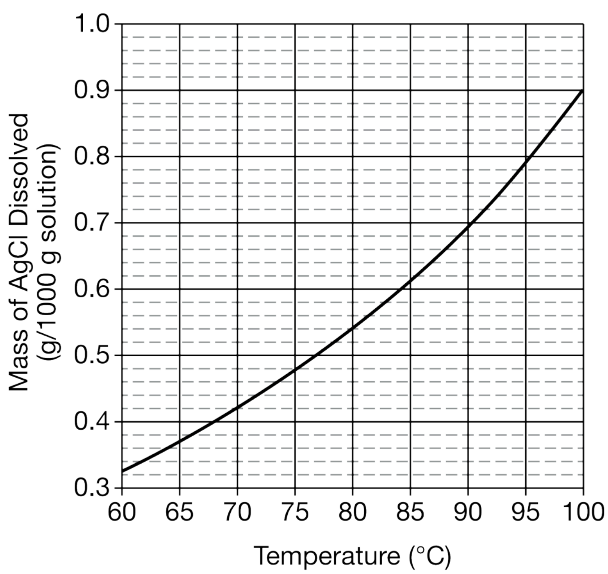
(f) The student finds the following particulate model of $AgCl(s)$. Assuming the crystal structures are similar, how should the student modify the model to represent $AgI(s)$? Justify your answer in terms of the radius and arrangement of electrons in the ions.

## Chapter 10 Part 3: Ksp




(g) The enthalpy of dissolution of $\mathrm{AgCl}(s)$ in water is a positive value. Based on this information, the student claims that the energy required to separate the ions in $\mathrm{AgCl}(s)$ is greater than the energy released during the hydration of $\mathrm{Ag}^{+}$ and $\mathrm{Cl}^{-}$ by water molecules. Do you agree or disagree with the student's claim? Justify your answer.

The student then conducts an experiment to determine the solubility of $\mathrm{AgCl}$ in a $2.5\ M$ $\mathrm{NaCl}$ solution at various temperatures. Some $\mathrm{AgCl}(s)$ remains in the container at all temperatures studied. The data are plotted on the graph below.




(h) A $1000.$ g sample of a saturated solution at $100.^{\circ}\mathrm{C}$ is cooled to $60.^{\circ}\mathrm{C}$. Determine the mass of $\mathrm{AgCl}(s)$ that would precipitate out of the solution.

(i) The specific heat capacity of the solution is $3.4\ \mathrm{J}/(\mathrm{g}\cdot{}^{\circ}\mathrm{C})$ . How much heat is released when $1000.$ g of the solution is cooled from $100.^{\circ}\mathrm{C}$ to $60.^{\circ}\mathrm{C}$ ?

# Chapter 10 Part 3: Ksp

$$\begin{array}{lll} AgCl(s) \rightleftarrows Ag^+(aq) + Cl^-(aq) & & K_{sp} = 1.0 \times 10^{-10} \\ Ag^+(aq) + 2\,Cl^-(aq) \rightleftarrows [AgCl_2]^-(aq) & & K_2 = 1.8 \times 10^5 \\ \hline AgCl(s) + Cl^-(aq) \rightleftarrows [AgCl_2]^-(aq) & & K_3 = ? \end{array}$$

(j) The student finds that $\mathrm{AgCl}$ is more soluble in $\mathrm{NaCl}(aq)$ than in water because of the series of reactions represented above. Determine the value of $K_3$ for the overall reaction.

To further investigate the reactivity of silver, the student submerges a piece of metallic nickel, $\mathrm{Ni}(s)$, in a solution of $\mathrm{AgNO_3}(aq)$ and observes that the following reaction occurs.

$\mathrm{Ni}(s) + 2\,\mathrm{AgNO_3}(aq) \rightarrow 2\,\mathrm{Ag}(s) + \mathrm{Ni(NO_3)_2}(aq)$

(k) Write the balanced net-ionic equation for the reaction.

(l) The mass of the $\mathrm{Ni}(s)$ decreases significantly as it sits in the $\mathrm{AgNO_3}$ solution. The student claims that the mass decreases because the atoms in $\mathrm{Ni}(s)$ lose electrons as the experiment progresses. Does the student's claim completely explain the decrease in the mass of the $\mathrm{Ni}(s)$? Justify your answer.

**Part A**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 ✓ |
|---|---|

The response shows a correct calculation of moles.

· $3.578\ \mathrm{g} - 0.718\ \mathrm{g} = 2.860\ \mathrm{g\ AgCl}(s)$

$2.860\,\mathrm{g\,AgCl}(s) \times \frac{1\,\mathrm{mol\,AgCl}(s)}{143\,\mathrm{g\,AgCl}(s)} = 2.00 \times 10^{-2}\,\mathrm{mol\,AgCl}(s)$

**Part B**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 ✓ |
|---|---|

The response shows a correct calculation of molarity.

· $\frac{2.00\times10^{-2}\,\mathrm{mol}}{0.1000\,\mathrm{L}} = 0.200\,M$

## Chapter 10 Part 3: Ksp

### Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides both the correct choice and a valid justification equivalent to the following.

· Disagree. The extra mass from the moisture would be calculated as additional $\mathrm{AgCl}(s)$. This error would lead to a higher calculated number of moles of $\mathrm{AgCl}(s)$ and therefore a higher concentration of $\mathrm{Ag}^{+}(aq)$ in the original solution.

### Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response shows a correct calculation of concentration.

· $K_{sp} = 1.0 \times 10^{-10} = [\mathrm{Ag}^{+}][\mathrm{Cl}^{-}] = x^2$

$[\mathrm{Ag}^{+}] = x = \sqrt{1.0 \times 10^{-10}} M = 1.0 \times 10^{-5} M$

### Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides both the correct answer and a valid justification equivalent to one of the following.

· The $\mathrm{AgCl}$ solution will have the greater $[\mathrm{Ag}^{+}]$. $\mathrm{AgCl}$ has a larger value of $K_{sp}$ than $\mathrm{AgI}$, indicating a higher concentration of each ion in a saturated solution since both compounds have $1 : 1$ ion stoichiometry.

OR

· The $\mathrm{AgCl}$ solution will have the greater $[\mathrm{Ag}^{+}]$. For a saturated solution of $\mathrm{AgI}$, $[\mathrm{Ag}^{+}] = x = \sqrt{8.3 \times 10^{-17}} M = 9.1 \times 10^{-9} M$, which is less than the $[\mathrm{Ag}^{+}]$ for a saturated solution of $\mathrm{AgCl}$.

### Part F

## Chapter 10 Part 3: Ksp

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response meets both of the following criteria.

- The response indicates that the $I^-$ ions should be larger.
- The response provides the explanation that $I^-$ has a larger ionic radius than $Cl^-$ due to more occupied electron shells ($n = 5$ versus $n = 3$).

**Part G**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides both the correct choice and a valid justification equivalent to the following.

· Agree. Since the enthalpy of solution is positive, more energy is absorbed by the system to overcome ion-ion attractions than is released when the ions were hydrated by water molecules.

**Part H**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides a correct calculation of mass.

· $0.90 \text{ g} - 0.32 \text{ g} = 0.58 \text{ g}$

**Part I**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

## Chapter 10 Part 3: Ksp

The response provides a correct calculation of the heat released.

$\cdot\, q = mc\Delta T = (1000.\,\mathrm{g})(3.4\,\frac{\mathrm{J}}{(\mathrm{g}\cdot{}^{\circ}\mathrm{C})})(60\,{}^{\circ}\mathrm{C} - 100\,{}^{\circ}\mathrm{C}) = -140{,}000\,\mathrm{J} = -140\,\mathrm{kJ}$

**Part J**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides the correct value.

$\cdot\, K_3 = \frac{[[\mathrm{AgCl_2}]^-]}{[\mathrm{Cl^-}]} = (K_{sp})(K_2) = (1.0 \times 10^{-10})(1.8 \times 10^{5}) = 1.8 \times 10^{-5}$

**Part K**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides the correctly balanced net-ionic equation.

$\cdot\, \mathrm{Ni}(s) + 2\,\mathrm{Ag}^{+}(aq) \rightarrow 2\,\mathrm{Ag}(s) + \mathrm{Ni}^{2+}(aq)$

(State symbols are not required.)

**Part L**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides both the correct answer and a valid justification equivalent to the following.

· No. Although $\mathrm{Ni}(s)$ does lose electrons, their mass is insignificant compared to the mass of the atoms. The mass of the $\mathrm{Ni}(s)$ decreases because it is converted into $\mathrm{Ni}^{2+}$ ions that migrate away from the metal as they disperse in the solution.

# Chapter 10 Part 3: Ksp

**10.** **Directions: For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.**

A student investigates various properties of silver compounds in the lab. First, the student carries out the following procedure to determine the concentration of $Ag^{+}(aq)$ in a solution of $AgNO_3(aq)$. The student adds an excess of $4.0\ M$ $HCl(aq)$ to $400.0\ mL$ of the solution, causing $AgCl(s)$ to precipitate. The precipitate is separated from the solution by filtering it through a pre-weighed piece of filter paper. The precipitate and filter paper are rinsed with distilled water, dried, and weighed. The data from the experiment are given in the table below.

| | |
|---|---|
| Initial mass of filter paper | $0.821\ g$ |
| Mass of filter paper +$AgCl(s)$ precipitate after drying | $1.110\ g$ |

(a) Calculate the number of moles of $AgCl(s)$ that precipitated. (The molar mass of $AgCl$ is $143\ g/mol$).

(b) Assuming that all of the $Ag^{+}(aq)$ ions precipitated, calculate the original molarity of $Ag^{+}(aq)$ in the solution before $HCl(aq)$ was added.

(c) The calculated concentration of $Ag^{+}(aq)$ from the student's data is higher than the actual concentration of the original solution. The student claims that the calculated concentration is too high because the filter paper and $AgCl(s)$ precipitate were not fully dry. Do you agree or disagree? Justify your answer.

After the experiment, the student learns that the precipitate, $AgCl(s)$, is slightly soluble in water. To determine the solubility, the student mixes a sample of the precipitate with warm, distilled water to make a saturated solution. The student finds the following equation for the dissolution of $AgCl(s)$ in water and a $K_{sp}$ value at the temperature of the water.

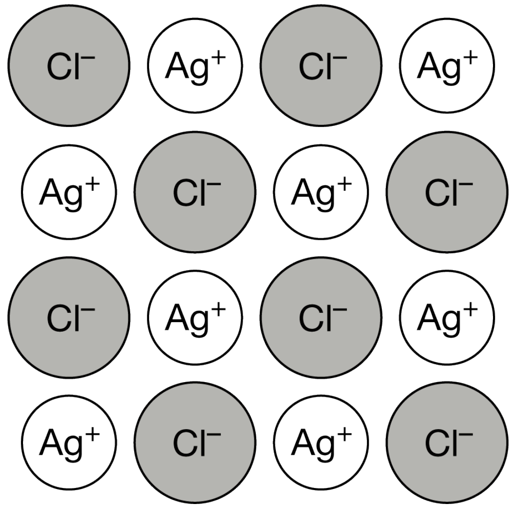
$$AgCl(s) \rightleftarrows Ag^{+}(aq) + Cl^{-}(aq) \qquad K_{sp} = 1.6 \times 10^{-9}$$

(d) Calculate $[Cl^{-}]$ in the student's solution.

(e) The value of $K_{sp}$ of another chloride salt, $CuCl(s)$, is $1.7 \times 10^{-7}$. If separate, saturated solutions of $CuCl$ and $AgCl$ are prepared, which solution will have the greater $[Cl^{-}]$? Justify your answer.

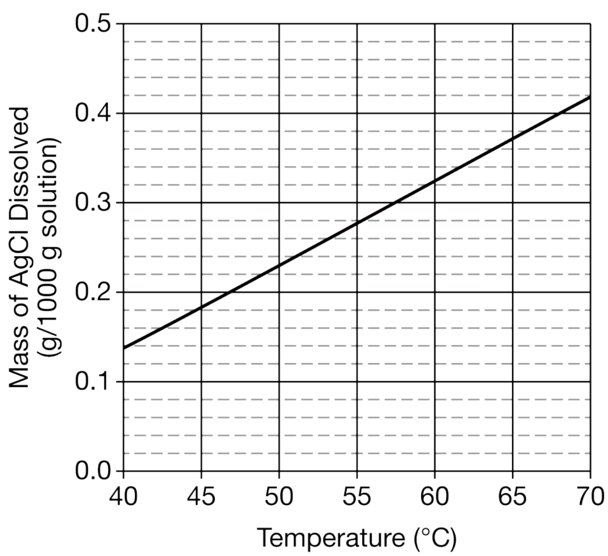
(f) The student finds the following particulate model of $AgCl(s)$. Assuming the crystal structures are similar, how should the student modify the model to represent $CuCl(s)$? Justify your answer in terms of the radius and arrangement of electrons in the ions.

**Chapter 10 Part 3: Ksp**




(g) The enthalpy of dissolution of $\mathrm{AgCl}(s)$ in water is a positive value. Based on this information, the student claims that the energy required to separate the ions in $\mathrm{AgCl}(s)$ is greater than the energy absorbed during the hydration of $\mathrm{Ag}^{+}$ and $\mathrm{Cl}^{-}$ by water molecules. Do you agree or disagree with the student's claim? Justify your answer.

The student then conducts an experiment to determine the solubility of $\mathrm{AgCl}$ in a $2.5\ M\ \mathrm{NaCl}$ solution at various temperatures. Some $\mathrm{AgCl}(s)$ remains in the container at all temperatures studied. The data are plotted on the graph below.




(h) A $1000.$ g sample of a saturated solution at $50.^{\circ}\mathrm{C}$ is heated to $70.^{\circ}\mathrm{C}$. Determine the additional mass of $\mathrm{AgCl}(s)$ that would dissolve into the solution.

(i) The specific heat capacity of the solution is $3.3\ \mathrm{J/(g\cdot{}^{\circ}C)}$ . How much heat is absorbed when $1000.$ g of the solution is heated from $50.^{\circ}\mathrm{C}$ to $70.^{\circ}\mathrm{C}$ ?

## Chapter 10 Part 3: Ksp

$$\begin{array}{rcll} AgCl(s) & \rightleftarrows & Ag^+(aq) + Cl^-(aq) & K_{sp} = 1.6 \times 10^{-9} \\ Ag^+(aq) + 2\,Cl^-(aq) & \rightleftarrows & [AgCl_2]^-(aq) & K_2 = 1.0 \times 10^5 \\ \hline AgCl(s) + Cl^-(aq) & \rightleftarrows & [AgCl_2]^-(aq) & K_3 = ? \end{array}$$

(j) The student finds that $\mathrm{AgCl}$ is more soluble in $\mathrm{NaCl}(aq)$ than in water because of the series of reactions represented above. Determine the value of $K_3$ for the overall reaction.

To further investigate the reactivity of silver, the student submerges a piece of metallic vanadium, $\mathrm{V}(s)$, in a solution of $\mathrm{AgNO_3}(aq)$ and observes that the following reaction occurs.

$\mathrm{V}(s) + 2\,\mathrm{AgNO_3}(aq) \rightarrow 2\,\mathrm{Ag}(s) + \mathrm{V(NO_3)_2}(aq)$

(k) Write the balanced net-ionic equation for the reaction.

(l) The mass of the $\mathrm{V}(s)$ decreases significantly as it sits in the $\mathrm{AgNO_3}$ solution. The student claims that the mass decreases because the atoms in the $\mathrm{V}(s)$ lose protons as the experiment progresses. Does the student's claim accurately explain the decrease in the mass of the $\mathrm{V}(s)$? Justify your answer.

**Part A**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response shows a correct calculation of moles.

· $1.110\text{ g} - 0.821\text{ g} = 0.289\text{ g AgCl}(s)$ $0.289\,\mathrm{g\,AgCl}(s) \times \frac{1\,\mathrm{mol\,AgCl}(s)}{143\,\mathrm{g\,AgCl}(s)} = 2.02 \times 10^{-3}\,\mathrm{mol}$

**Part B**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response shows a correct calculation of molarity.

· $\frac{2.02\times10^{-3}\,\mathrm{mol}}{0.4000\,\mathrm{L}} = 5.05 \times 10^{-3}\,M$

**Part C**

## Chapter 10 Part 3: Ksp

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
| --- | --- |

The response provides both the correct choice and a valid justification equivalent to the following.

· Agree. The extra mass from the moisture would be calculated as additional $\mathrm{AgCl}(s)$. This error would lead to a higher calculated number of moles of $\mathrm{AgCl}$ and therefore a higher concentration of $\mathrm{Ag}^+(aq)$ in the original solution.

**Part D**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
| --- | --- |

The response shows a correct calculation of concentration.

· $1.6 \times 10^{-9} = [\mathrm{Ag}^+][\mathrm{Cl}^-] = x^2$

$[\mathrm{Cl}^-] = x = \sqrt{1.6 \times 10^{-9}}M = \sqrt{16 \times 10^{-10}}M = 4.0 \times 10^{-5}M$

**Part E**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
| --- | --- |

The response provides both the correct answer and a valid justification equivalent to the following.

· The $\mathrm{CuCl}$ solution will contain a greater number of moles of $\mathrm{Cl}^-$ ions. $\mathrm{CuCl}$ has a larger value of $K_{sp}$ than $\mathrm{AgCl}$, indicating a higher concentration of each ion in a saturated solution since both compounds have $1 : 1$ ion stoichiometry.

**Part F**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
| --- | --- |

## Chapter 10 Part 3: Ksp

The response meets both of the following criteria.

- The response indicates that the $Cu^+$ ions should be smaller.
- The response provides the explanation that $Cu^+$ has a smaller ionic radius than $Ag^+$ due to fewer occupied electron shells ($n = 3$ versus $n = 4$).

### Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides both the correct choice and a valid justification equivalent to the following.

· Disagree. The hydration of the ions releases energy. Energy is not absorbed during this process.

### Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides a correct calculation of mass.

· $0.42 \text{ g} - 0.23 \text{ g} = 0.19 \text{ g}$

### Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides a correct calculation of the heat absorbed.

· $q = mc\Delta T = (1000.\text{ g})(3.3\,\frac{\text{J}}{(\text{g}\cdot{}^\circ\text{C})})(70.\,{}^\circ\text{C} - 50.\,{}^\circ\text{C}) = 66,000\,\text{J} = 66\,\text{kJ}$

### Part J

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

## Chapter 10 Part 3: Ksp

| 0 | 1 |
|---|---|

The response provides the correct value.

· $K_3 = \frac{[[AgCl_2]^-]}{[Cl^-]} = (K_{sp})(K_2) = (1.6 \times 10^{-9})(1.0 \times 10^5) = 1.6 \times 10^{-4}$

**Part K**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides the correctly balanced net-ionic equation.

· $V(s) + 2\,Ag^+(aq) \rightarrow 2\,Ag(s) + V^{2+}(aq)$

(State symbols are not required.)

**Part L**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides both the correct answer and a valid justification equivalent to the following.

· No. Electrons, not protons, are exchanged in the oxidation of $V(s)$.

## Chapter 10 Part 3: Ksp

**11.** **Directions: For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.**

A student investigates various properties of silver compounds in the lab. First, the student carries out the following procedure to determine the concentration of $Ag^+(aq)$ in a solution of $AgNO_3(aq)$. The student adds an excess of $6.0\ M\ HCl(aq)$ to $500.0\ mL$ of the solution, causing $AgCl(s)$ to precipitate. The precipitate is separated from the solution by filtering it through a pre-weighed piece of filter paper. The precipitate and filter paper are rinsed with distilled water, dried, and weighed. The data from the experiment are given in the table below.

| | |
|---|---|
| Initial mass of filter paper | $0.945\ g$ |
| Mass of filter paper +$AgCl(s)$precipitate after drying | $3.095\ g$ |

(a) Calculate the number of moles of $AgCl(s)$ that precipitated. (The molar mass of $AgCl$ is $143\ g/mol$).

(b) Assuming that all of the $Ag^+(aq)$ ions precipitated, calculate the original molarity of $Ag^+(aq)$ in the solution before $HCl(aq)$ was added.

(c) The calculated concentration of $Ag^+(aq)$ from the student's data is higher than the actual concentration of the original solution. The student claims that the calculated concentration is too high because $8.0\ M\ HCl(aq)$ was accidentally used instead of $6.0\ M\ HCl(aq)$. Do you agree or disagree? Justify your answer.

After the experiment, the student learns that the precipitate, $AgCl(s)$, is slightly soluble in water. To determine the solubility, the student mixes a sample of the precipitate with warm, distilled water to make a saturated solution. The student finds the following equation for the dissolution of $AgCl(s)$ in water and a $K_{sp}$ value at the temperature of the water.

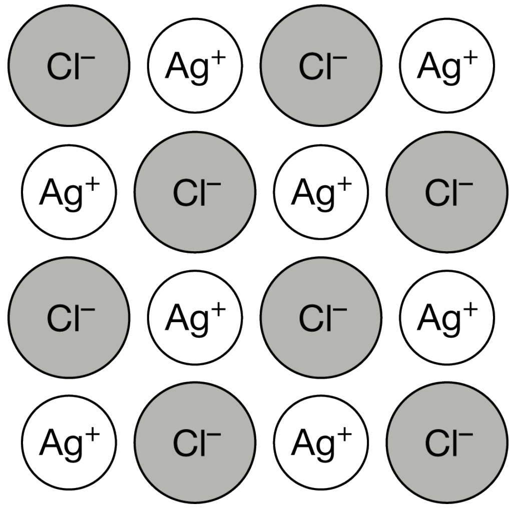
$$AgCl(s) \rightleftarrows Ag^+(aq) + Cl^-(aq) \qquad K_{sp} = 2.5 \times 10^{-9}$$

(d) Calculate $[Cl^-]$ in the student's solution.

(e) The value of $K_{sp}$ of another chloride salt, $LiCl(s)$, is $4.0 \times 10^2$. If separate, saturated solutions of $LiCl$ and $AgCl$ are prepared, which solution will have the greater $[Cl^-]$? Justify your answer.

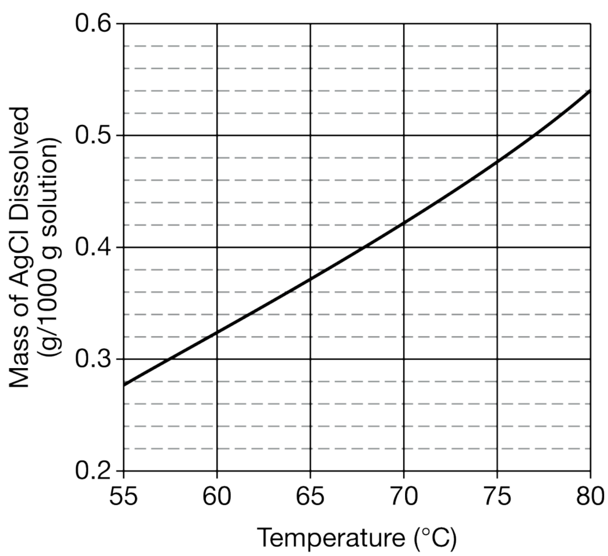
(f) The student finds the following particulate model of $AgCl(s)$. Assuming the crystal structures are similar, how should the student modify the model to represent $LiCl(s)$? Justify your answer in terms of the radius and arrangement of electrons in the ions.

## Chapter 10 Part 3: Ksp




(g) The enthalpy of dissolution of $\mathrm{AgCl}(s)$ in water is a positive value. Based on this information, the student claims that the energy released during the separation of the ions in $\mathrm{AgCl}(s)$is greater than the energy absorbed during the hydration of $\mathrm{Ag}^{+}$ and $\mathrm{Cl}^{-}$ by water molecules. Do you agree or disagree with the student's claim? Justify your answer.

The student then conducts an experiment to determine the solubility of $\mathrm{AgCl}$ in a $2.5\ M$ $\mathrm{NaCl}$ solution at various temperatures. Some $\mathrm{AgCl}(s)$ remains in the container at all temperatures studied. The data are plotted on the graph below.




(h) A $1000.$ g sample of a saturated solution at $60.^{\circ}\mathrm{C}$ is heated to $75^{\circ}\mathrm{C}$. Determine the additional mass of $\mathrm{AgCl}(s)$ that would dissolve into the solution.

(i) The specific heat capacity of the solution is $3.4\ \mathrm{J/(g \cdot {}^{\circ}C)}$ . How much heat is absorbed when $1000.$ g of the solution is heated from $60.^{\circ}\mathrm{C}$ to $75^{\circ}\mathrm{C}$ ?

## Chapter 10 Part 3: Ksp

$$
\begin{array}{rcll}
\mathrm{AgCl}(s) & \rightleftarrows & \mathrm{Ag^+}(aq) + \mathrm{Cl^-}(aq) & K_{sp} = 2.5 \times 10^{-9} \\
\mathrm{Ag^+}(aq) + 2\,\mathrm{Cl^-}(aq) & \rightleftarrows & [\mathrm{AgCl_2}]^-(aq) & K_2 = 2.0 \times 10^{5} \\
\hline
\mathrm{AgCl}(s) + \mathrm{Cl^-}(aq) & \rightleftarrows & [\mathrm{AgCl_2}]^-(aq) & K_3 = ?
\end{array}
$$

(j) The student finds that $\mathrm{AgCl}$ is more soluble in $\mathrm{NaCl}(aq)$ than in water because of the series of reactions above. Determine the value of $K_3$ for the overall reaction.

To further investigate the reactivity of silver, the student submerges a piece of metallic zirconium, $\mathrm{Zr}(s)$, in a solution of $\mathrm{AgNO_3}(aq)$ and observes that the following reaction occurs.

$\mathrm{Zr}(s) + 4\,\mathrm{AgNO_3}(aq) \rightarrow 4\,\mathrm{Ag}(s) + \mathrm{Zr(NO_3)_4}(aq)$

(k) Write the balanced net-ionic equation for the reaction.

(l) The mass of the $\mathrm{Zr}(s)$ decreases significantly as it sits in the $\mathrm{AgNO_3}$ solution. The student claims that the mass decreases because the $\mathrm{Zr}(s)$ melts into a liquid as the experiment progresses. Does the student's claim accurately explain the decrease in the mass of the $\mathrm{Zr}(s)$? Justify your answer.

**Part A**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
| --- | --- |

The response shows a correct calculation of moles.

- $3.095\text{ g} - 0.945\text{ g} = 2.150\text{ g AgCl}(s)$ $2.150\,\text{g AgCl}(s) \times \frac{1\,\text{mol AgCl}(s)}{143\,\text{g AgCl}(s)} = 1.50 \times 10^{-2}\,\text{mol}$

**Part B**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
| --- | --- |

The response shows a correct calculation of molarity.

- $\frac{1.50 \times 10^{-2}\,\text{mol}}{0.5000\,\text{L}} = 3.00 \times 10^{-2}\,M$

**Part C**

## Chapter 10 Part 3: Ksp

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides the correct choice and a valid justification equivalent to the following.

· Disagree. The $HCl(aq)$ reactant is in excess, so using a higher concentration would not affect the amount of precipitate that is formed.

**Part D**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response shows a correct calculation of concentration.

· $K_{sp} = 2.5 \times 10^{-9} = [Ag^{+}][Cl^{-}] = x^2$

$[Cl^{-}] = x = \sqrt{2.5 \times 10^{-9}\ M} = \sqrt{25 \times 10^{-10} M} = 5.0 \times 10^{-5} M$

**Part E**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides both the correct answer and a valid justification equivalent to one of the following.

· The $LiCl$ solution will have the greater $[Cl^{-}]$. $LiCl$ has a larger value of $K_{sp}$ than $AgCl$, indicating a higher concentration of each ion in a saturated solution since both compounds have a $1\ :\ 1$ ion stoichiometry.

OR

· The $LiCl$ solution will have the greater $[Cl^{-}]$. For a saturated solution of $LiCl$, $[Cl^{-}] = x = \sqrt{4.0 \times 10^{2}\ M} = 2.0 \times 10^{1} M$, which is greater than the $[Cl^{-}]$ for a saturated solution of $AgCl$.

**Part F**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

## Chapter 10 Part 3: Ksp

| 0 | 1 |
|---|---|

The response meets both of the following criteria.

- The response indicates that the $Li^+$ ions should be smaller.
- The response provides the explanation that $Li^+$ has a smaller ionic radius than $Ag^+$ due to fewer occupied electron shells ($n = 2$ versus $n = 4$).

**Part G**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides the correct choice and a valid justification equivalent to the following.

· Disagree. Energy is absorbed, not released, in separating the ions in $AgCl$. Also, the hydration of the ions releases energy and does not absorb energy. Both of the student's claims are incorrect.

**Part H**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides a correct calculation of mass.

· $0.48\text{ g} - 0.32\text{ g} = 0.16\text{ g}$

**Part I**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides a correct calculation of the heat absorbed.

## Chapter 10 Part 3: Ksp

· $q = mc\Delta T = (1000.\,\mathrm{g})(3.4\,\frac{\mathrm{J}}{(\mathrm{g}\cdot{}^{\circ}\mathrm{C})})(75\,{}^{\circ}\mathrm{C} - 60\,{}^{\circ}\mathrm{C}) = 51,000\,\mathrm{J} = 51\,\mathrm{kJ}$

**Part J**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 ✓ |
|---|---|

The response provides the correct value.

· $K_3 = \frac{[[\mathrm{AgCl_2}]^-]}{[\mathrm{Cl^-}]} = (K_{sp})(K_2) = (2.5 \times 10^{-9})(2.0 \times 10^5) = 5.0 \times 10^{-4}$

**Part K**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 ✓ |
|---|---|

The response provides the correctly balanced net-ionic equation.

· $\mathrm{Zr}(s) + 4\,\mathrm{Ag^+}(aq) \rightarrow 4\,\mathrm{Ag}(s) + \mathrm{Zr^{4+}}(aq)$

(State symbols are not required.)

**Part L**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 ✓ |
|---|---|

The response provides both the correct answer and a valid justification equivalent to the following.

· No. The mass loss comes from the oxidation of $\mathrm{Zr}(s)$ into $\mathrm{Zr^{4+}}(aq)$ ions that disperse in the solution.

## Chapter 10 Part 3: Ksp

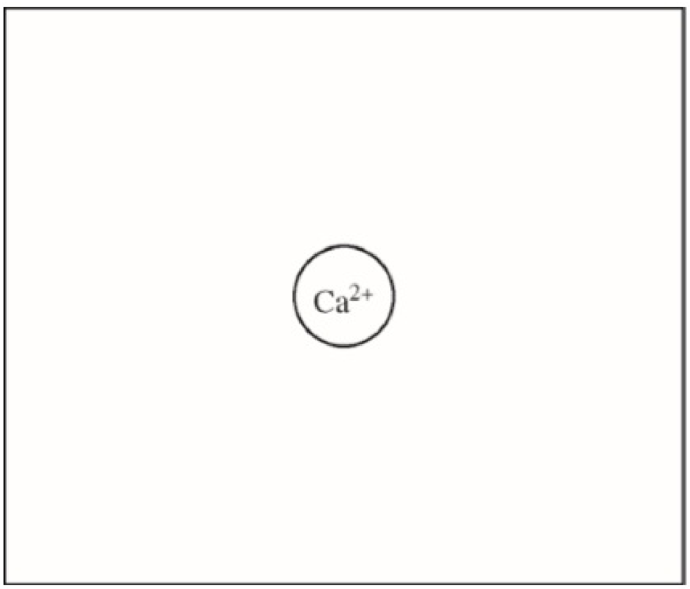
**12.** Answer the following questions about the solubility of $Ca(OH)_2$ ($K_{sp}$ = 1.3 x $10^{-6}$ ) .

a. Write a balanced chemical equation for the dissolution of $Ca(OH)_2(s)$ in pure water.

b. Calculate the molar solubility of $Ca(OH)_2$ in 0.10 $M$ $Ca(NO_3)_2$ .

c. In the box below, complete a particle representation diagram that includes four water molecules with proper
orientation around the $Ca^{2+}$ ion.

Represent water molecules as  .




**Part A**

1 point is earned for the correct equation.

Ca(OH)2 ⇄ Ca2+ + 2 OH-

| 0 | 1 |
|---|---|

The student response earns one of the following points:

1 point is earned for the correct equation.

Ca(OH)2 ⇄ Ca2+ + 2 OH-

## Chapter 10 Part 3: Ksp

**Part B**

1 point is earned for the correct stoichiometry and setup.

$K_{sp} = [Ca^{2+}]\,[OH^-]^2$

$1.3 \times 10^{-6} = (0.10 + x)\,(2x)^2 \approx (0.10)\,4x^2$ [assuming $x << 0.10$]

$1.3 \times 10^{-5} = 4x^2$

$x = 0.0018\ M$

1 point is earned for the final answer.

Molar solubility of $Ca(OH)_2 = 0.0018\ M$

| | | |
|---|---|---|
| 0 | 1 | 2 ✓ |

The student response earns two of the following points:

1 point is earned for the correct stoichiometry and setup.

$K_{sp} = [Ca^{2+}]\,[OH^-]^2$

$1.3 \times 10^{-6} = (0.10 + x)\,(2x)^2 \approx (0.10)\,4x^2$ [assuming $x << 0.10$]

$1.3 \times 10^{-5} = 4x^2$

$x = 0.0018\ M$

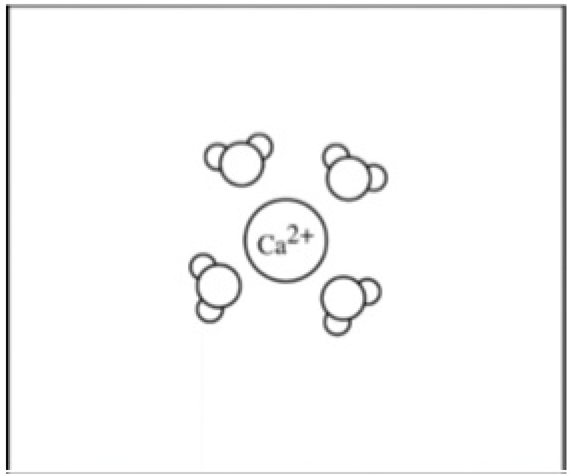
1 point is earned for the final answer.

Molar solubility of $Ca(OH)_2 = 0.0018\ M$

**Part C**

1 point is earned for a correct diagram that shows at least three of the four water molecules oriented as described.

# Chapter 10 Part 3: Ksp

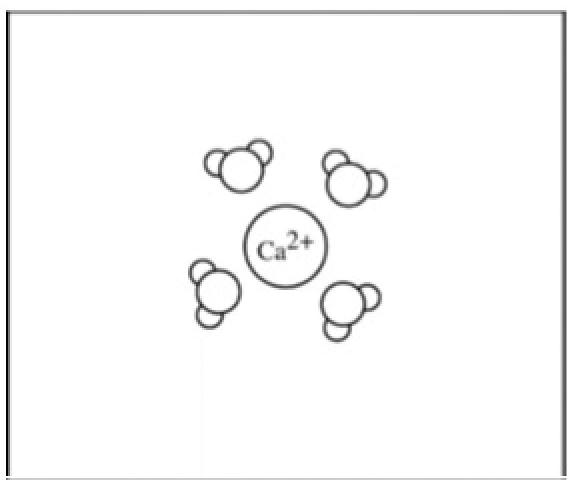



[The diagram should show the oxygen side of the water molecules oriented closer to the Ca2+ ion.]

| 0 | 1 |
|---|---|

The student response earns one of the following points:

1 point is earned for a correct diagram that shows at least three of the four water molecules oriented as described.




[The diagram should show the oxygen side of the water molecules oriented closer to the Ca2+ ion.]

## Chapter 10 Part 3: Ksp

**13.** Answer the following questions about the solubility of AgCl(*s*). The value of $K_{sp}$ for AgCl(*s*) is $1.8 \times 10^{-10}$.

a. Calculate the value of $[Ag^+]$ in a saturated solution of AgCl in distilled water.

b. The concentration of $Cl^-(aq)$ in seawater is 0.54 *M*.

i. Calculate the molar solubility of AgCl(*s*) in seawater.

ii. Explain why AgCl(*s*) is less soluble in seawater than in distilled water.

**Part A**

**2 point(s) maximum**

1 point is earned for the correct Ksp expression and indication that the two ions have equal concentrations.

Ksp = [Ag+][Cl-]

Let x = [Ag+] = [Cl-]

1 point is earned for correct calculation of the value of [Ag+].

$$\text{Then } 1.8 \times 10^{-10} = (x)(x) \Rightarrow x = \sqrt{1.8 \times 10^{-10}}$$

x = [Ag+] = 1.3 x 10-5 *M*

| 0 | 1 | 2 |
|---|---|---|

**Student response earns 0 of the following 2 point(s)**

**2 point(s) maximum**

1 point is earned for the correct Ksp expression and indication that the two ions have equal concentrations.

Ksp = [Ag+][Cl-]

Let x = [Ag+] = [Cl-]

1 point is earned for correct calculation of the value of [Ag+].

$$\text{Then } 1.8 \times 10^{-10} = (x)(x) \Rightarrow x = \sqrt{1.8 \times 10^{-10}}$$

x = [Ag+] = 1.3 x 10-5 *M*

**Part B**

## Chapter 10 Part 3: Ksp

**2 point(s) maximum**

Part (i)

1 point is earned for the correct answer with supporting work.

1.8 x 10-10 = [Ag+] x (0.54) ⇒ [Ag+] = 3.3 x 10-10 *M*

Part (ii)

1 point is earned for a correct explanation

An increased [Cl- ] will decrease the solubility of AgCl(*s*) since the Ksp is a product of the [Ag+] and [Cl-]. (This is an example of the common ion effect.)

| 0 | 1 | 2 |
|---|---|---|

**Student response earns 2 of the following 2 point(s)**

**2 point(s) maximum**

Part (i)

1 point is earned for the correct answer with supporting work.

1.8 x 10-10 = [Ag+] x (0.54) ⇒ [Ag+] = 3.3 x 10-10 *M*

Part (ii)

1 point is earned for a correct explanation

An increased [Cl- ] will decrease the solubility of AgCl(*s*) since the Ksp is a product of the [Ag+] and [Cl-]. (This is an example of the common ion effect.)

---

Reaction 1: $CaC_2(s) + 2\ H_2O(l) \rightarrow C_2H_2(g) + Ca(OH)_2(s)$

Reaction 2: $NaOCl(aq) + 2\ HCl(aq) \rightarrow Cl_2(g) + NaCl(aq) + H_2O(l)$

Reaction 3: $C_2H_2(g) + Cl_2(g) \rightarrow C_2H_2Cl_2(g)$

**14.** $Ca(OH)_2(s)$ precipitates when a 1.0 g sample of $CaC_2(s)$ is added to 1.0 L of distilled water at room temperature. If a 0.064 g sample of $CaC_2(s)$ (molar mass 64 g/mol) is used instead and all of it reacts, which of the following will occur and why? (The value of $K_{sp}$ for $Ca(OH)_2$ is 8.0 x $10^{-8}$.)

## Chapter 10 Part 3: Ksp

(A) $Ca(OH)_2$ will precipitate because $Q > K_{sp}$ .

(B) $Ca(OH)_2$ will precipitate because $Q < K_{sp}$ .

(C) $Ca(OH)_2$ will not precipitate because $Q > K_{sp}$ .

(D) $Ca(OH)_2$ will not precipitate because $Q < K_{sp}$ . ✓

---

**15.** $CaF_2(s) \rightleftarrows Ca^{2+}(aq) + 2\,F^-(aq)$

$\Delta H > 0$

Dissolution of the slightly soluble salt $CaF_2$ is shown by the equation above. Which of the following changes will decrease $[Ca^{2+}]$ in a saturated solution of $CaF_2$ , and why? (Assume that after each change some $CaF_2(s)$ remains in contact with the solution.)

(A) Allowing some of the water to evaporate from the solution, because more $CaF_2(s)$ will precipitate

(B) Adding 0.1 $M$ $HNO_3(aq)$ , because some $F^-(aq)$ ions will become protonated

(C) Adding 0.1 $M$ $NaNO_3(aq)$ , because additional liquid will dilute the solution

(D) Adding $NaF(s)$, because the reaction will proceed toward reactants ✓

**16.**

$$CaF_2(s) \rightleftarrows Ca^{2+}(aq) + 2\,F^-(aq) \qquad K_{sp} = 4.0 \times 10^{-11}$$

The concentration of $F^-(aq)$ in drinking water that is considered to be ideal for promoting dental health is $4.0 \times 10^{-5}\ M$. Based on the information above, the maximum concentration of $Ca^{2+}(aq)$ that can be present in drinking water without lowering the concentration of $F^-(aq)$ below the ideal level is closest to

(A) $0.25\ M$

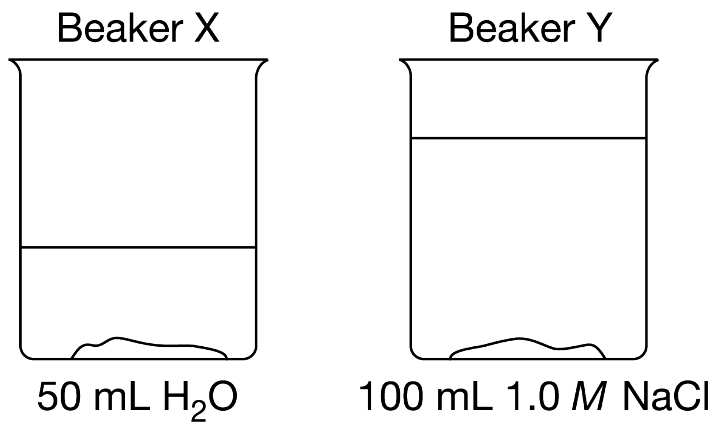
(B) $0.025\ M$ ✓

(C) $1.6 \times 10^{-6}\ M$

(D) $1.6 \times 10^{-15}\ M$

# Chapter 10 Part 3: Ksp

**17.**




Beaker $\mathrm{X}$ contains $50\ \mathrm{mL}$ of distilled water and beaker $\mathrm{Y}$ contains $100\ \mathrm{mL}$ of $1.0\ M\ \mathrm{NaCl}$. Solid $\mathrm{AgCl}$ is added to each of the beakers. After thoroughly stirring the contents of the beakers, some solid $\mathrm{AgCl}$ remains at the bottom of each beaker, as shown above. Which of the following is true?

(A) $[\mathrm{Ag}^+]$ is zero in both beakers.

(B) $[\mathrm{Ag}^+]$ is the same, but not zero, in both beakers.

(C) $[\mathrm{Ag}^+]$ is greater in beaker $\mathrm{X}$.

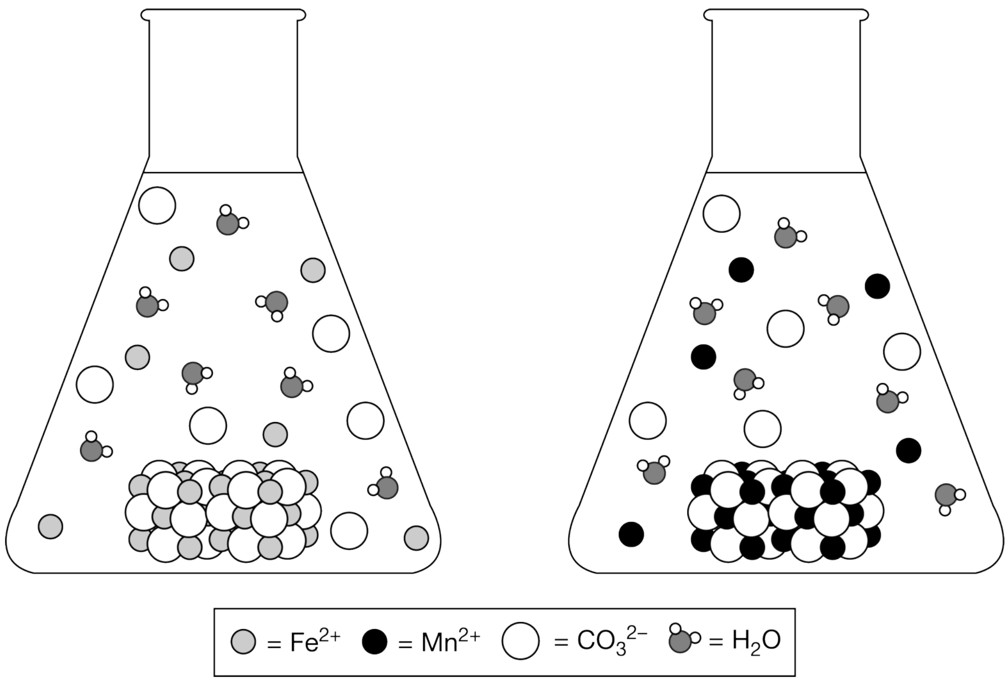
(D) $[\mathrm{Ag}^+]$ is greater in beaker $\mathrm{Y}$.

## Chapter 10 Part 3: Ksp

**18.**

| Reaction | $K_{sp}$ | $\Delta H^\circ$ | $\Delta S^\circ$ |
|---|---|---|---|
| $FeCO_3(s) \rightleftarrows Fe^{2+}(aq) + CO_3{}^{2-}(aq)$ | $3 \times 10^{-11}$ | $< 0$ | $> 0$ |
| $MnCO_3(s) \rightleftarrows Mn^{2+}(aq) + CO_3{}^{2-}(aq)$ | $2 \times 10^{-11}$ | $< 0$ | $> 0$ |

The table above lists the equilibrium constants and changes in thermodynamic properties for the dissolution of $FeCO_3$ and $MnCO_3$ at $25^\circ C$. The two particle diagrams below represent saturated solutions of each compound at equilibrium.




Which of the following explains which of the properties listed in the table is best represented by the particle diagram?

(A) The particle diagrams best represent that $\Delta H^\circ < 0$ because the ions from both compounds are solvated by water molecules.

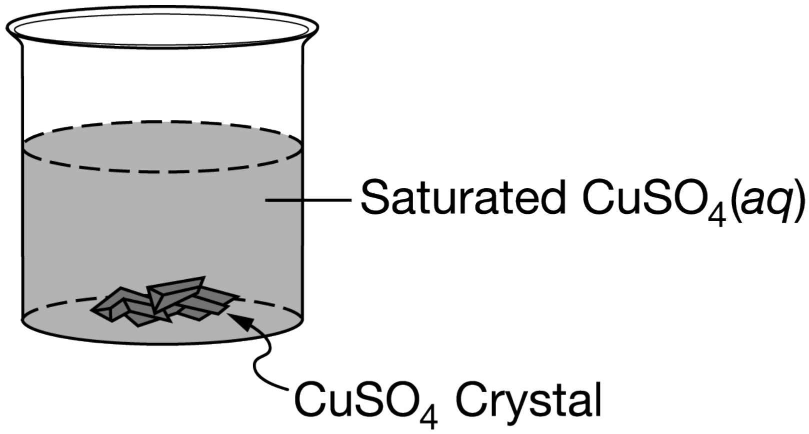
(B) The particle diagrams best represent that $\Delta H^\circ < 0$ because both compounds produce about the same amount of $CO_3{}^{2-}$ ions from the dissolution.

(C) The particle diagrams best represent that $\Delta S^\circ > 0$ because both compounds produce a very small amount of ions from the dissolution.

(D) The particle diagrams best represent that the molar solubility is greater for $FeCO_3$ compared to $MnCO_3$.

## Chapter 10 Part 3: Ksp

**19.**




The saturated $CuSO_4(aq)$ shown above is left uncovered on a lab bench at a constant temperature. As the solution evaporates, $1.0\ mL$ samples of the solution are removed every three days and the $[SO_4{}^{2-}]$ in the samples is measured. It is observed that the $[SO_4{}^{2-}]$ in the solution did not change over time. Which of the following best helps to explain the observation?

(A) As the solution evaporates, $Cu^{2+}(aq)$ and $SO_4{}^{2-}(aq)$ leave the beaker along with the water molecules.

(B) As the solution evaporates, the dissolution of $CuSO_4(s)$ in the beaker decreases.

(C) The evaporation of water is endothermic, so more $CuSO_4(s)$ dissolves exothermically in the solution, which increases the $[SO_4{}^{2-}]$.

(D) As water evaporates, more $CuSO_4(s)$ precipitates out of the solution in the beaker. ✓

**20.** $CdF_2(s) \rightleftarrows Cd^{2+}(aq) + 2\ F^-(aq)$

A saturated aqueous solution of $CdF_2$ is prepared. The equilibrium in the solution is represented above. In the solution, $[Cd^{2+}]_{eq} = 0.0585\ M$ and $[F^-]_{eq} = 0.117\ M$. Some $0.90\ M$ NaF is added to the saturated solution. Which of the following identifies the molar solubility of $CdF_2$ in pure water and explains the effect that the addition of NaF has on this solubility?

(A) The molar solubility of $CdF_2$ in pure water is $0.0585\ M$, and adding NaF decreases this solubility because the equilibrium shifts to favor the precipitation of some $CdF_2$. ✓

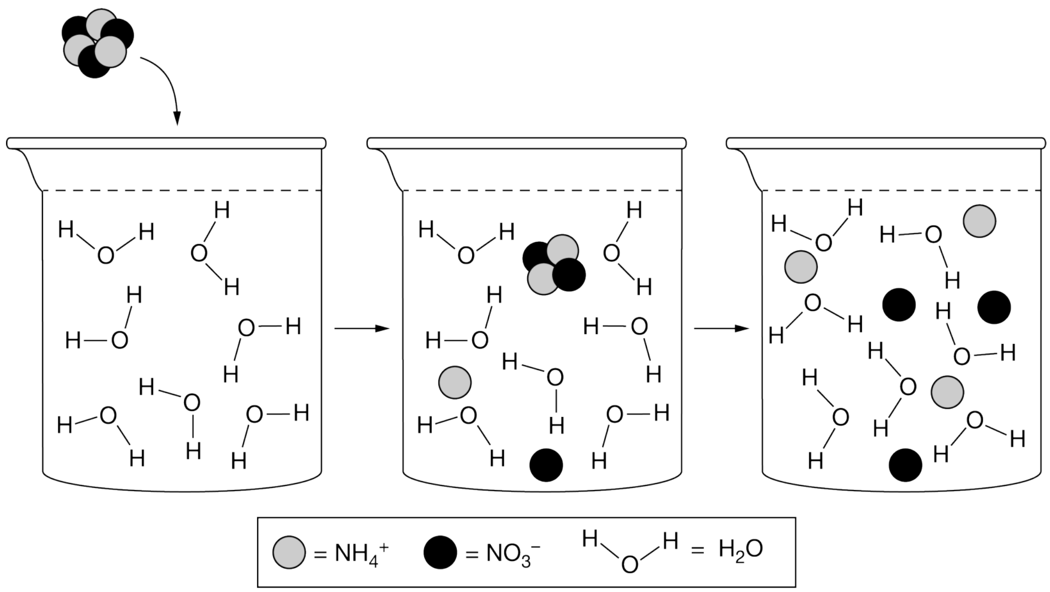
(B) The molar solubility of $CdF_2$ in pure water is $0.0585\ M$, and adding NaF has no effect on the solubility because only changes in temperature can increase or decrease the molar solubility of an ionic solid.

(C) The molar solubility of $CdF_2$ in pure water is $0.117\ M$, and adding NaF decreases this solubility because the equilibrium shifts to favor the precipitation of some $CdF_2$.

(D) The molar solubility of $CdF_2$ in pure water is $0.176\ M$, and adding NaF increases this solubility because the $Na^+$ ions displace the $Cd^{2+}$ ions, causing the equilibrium to shift to favor the products.

## Chapter 10 Part 3: Ksp

**21.**




At $298\ \mathrm{K}$, $\mathrm{NH_4NO_3}$ readily dissolves in water, suggesting that the change in free energy ($\Delta G$) favors the dissolution process. However, when $\mathrm{NH_4NO_3}$ dissolves in water, the temperature of the water decreases. The particulate diagram above attempts to provide a microscopic view of the dissolution of $\mathrm{NH_4NO_3}(s)$ considering both the change in enthalpy ($\Delta H$) and the change in entropy ($\Delta S$). Which of the following explains what the particle diagram is able to illustrate and why?

(A) The particle diagram is able to illustrate that the $\Delta H$ for the dissolution of $\mathrm{NH_4NO_3}$ is exothermic because overcoming the attractive forces between the ions requires the absorption of energy.

(B) The particle diagram is able to illustrate that the $\Delta H$ for the dissolution of $\mathrm{NH_4NO_3}$ is endothermic because forming new interactions between the ions and the water molecules releases energy.

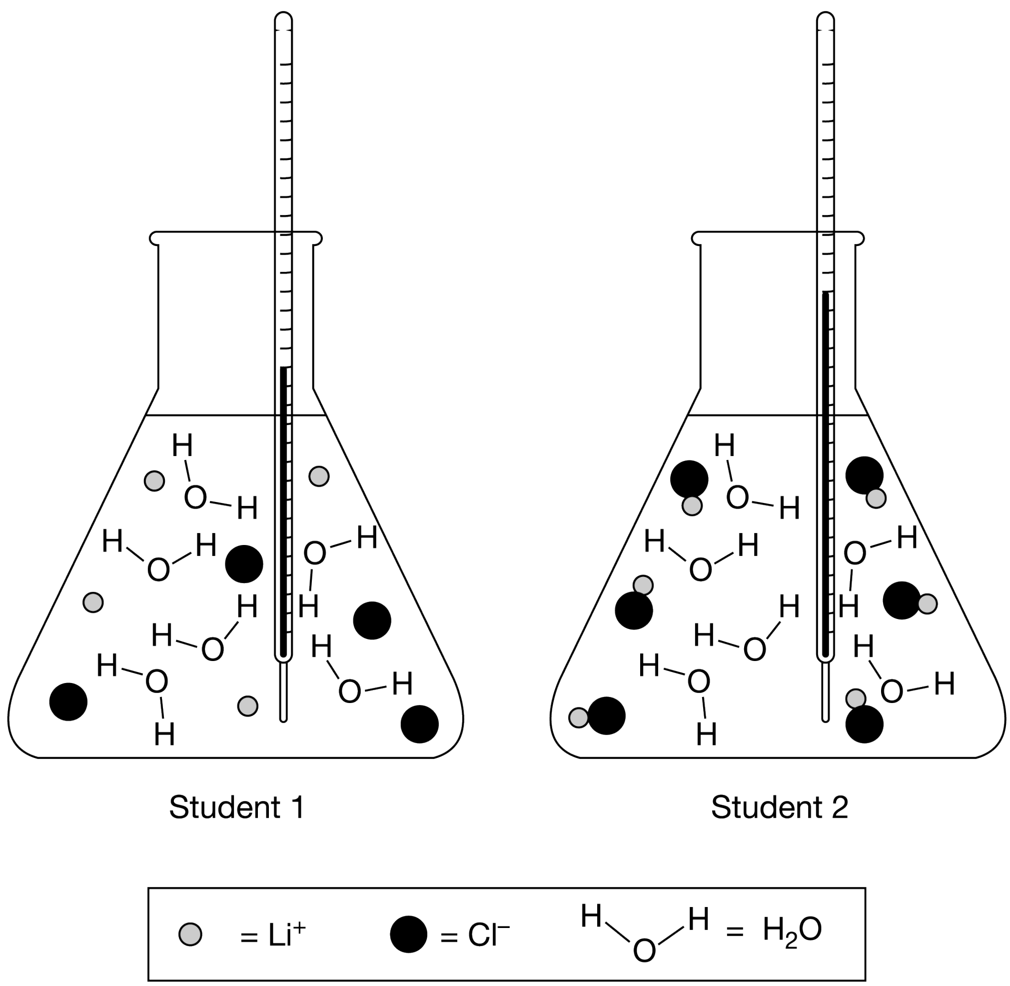
(C) The particle diagram is able to illustrate that entropy increases when $\mathrm{NH_4NO_3}(s)$ dissolves in water because the ions disperse in solution. ✓

(D) The particle diagram is able to illustrate that entropy decreases when $\mathrm{NH_4NO_3}(s)$ dissolves in water because many hydrogen bonds can form between the water molecules and the ions.

## Chapter 10 Part 3: Ksp

**22.**

| Property | Student 1 | Student 2 |
|---|---|---|
| Mass of $LiCl(g)$ | 2.0 | 3.0 |
| Mass of $H_2O(g)$ | 100.0 | 100.0 |
| Initial temperature of $H_2O(^\circ C)$ | 20.0 | 20.1 |
| Temperature of $H_2O$ after $LiCl$ dissolved $(^\circ C)$ | 23.9 | 26.1 |




Two students prepared aqueous solutions of $LiCl$ and measured the properties, as shown in the table above. Both students observed that the solid $LiCl$ readily dissolved in $H_2O$. The students drew particle diagrams to explain the changes in the enthalpy and entropy of dissolution for $LiCl$ based on their results and observations. Based on this information, the better particle diagram was drawn by which student, and why is that diagram more accurate?

## Chapter 10 Part 3: Ksp

(A) The better particle diagram was drawn by Student 1 because when $LiCl$ dissolves in water, it dissociates into $Li^{+}$ and $Cl^{-}$ ions causing an increase in entropy. ✓

(B) The better particle diagram was drawn by Student 1 because when $LiCl$ dissolves in water, it dissociates into $Li^{+}$ and $Cl^{-}$ ions indicating that the dissolution of $LiCl$ is exothermic.

(C) The better particle diagram was drawn by Student 2 because it shows that $LiCl$ does not dissociate, resulting in a decrease in entropy for the dissolution of $LiCl$.

(D) The better particle diagram was drawn by Student 2 because it shows that $LiCl$ does not dissociate, resulting in an endothermic process for the dissolution of $LiCl$.

**23.**

| Compound | $K_{sp}$ at 25°C |
|---|---|
| $Pb(OH)_2(s)$ | $1.2 \times 10^{-15}$ |
| $PbI_2(s)$ | $1.4 \times 10^{-8}$ |
| $PbF_2(s)$ | $4.0 \times 10^{-8}$ |

Equimolar samples of $Pb(OH)_2(s)$, $PbI_2(s)$, and $PbF_2(s)$ are placed in three separate beakers, each containing 250 mL of water at 25°C. After the solutions are stirred, solid remains in the bottom of each beaker. Based on the $K_{sp}$ values for the compounds listed in the table above, a solution of which of the compounds will have the lowest $[Pb^{2+}]$ ?

(A) $Pb(OH)_2(s)$ ✓

(B) $PbI_2(s)$ only

(C) $PbF_2(s)$ only

(D) Both $PbI_2(s)$ and $PbF_2(s)$ will produce solutions with the same, lowest $[Pb^{2+}]$.

**24.**

| Compound | $K_{sp}$ |
|---|---|
| $PbCl_2$ | $1.2 \times 10^{-5}$ |
| $CuCl$ | $1.6 \times 10^{-7}$ |
| $AgCl$ | $1.8 \times 10^{-10}$ |
| $Hg_2Cl_2$ | $1.4 \times 10^{-18}$ |

Based on the $K_{sp}$ values in the table above, a saturated solution of which of the following compounds has the highest $[Cl^{-}]$?

## Chapter 10 Part 3: Ksp

(A) $PbCl_2$ ✓
(B) $CuCl$
(C) $AgCl$
(D) $Hg_2Cl_2$

**25.** High solubility of an ionic solid in water is favored by which of the following conditions?

I. The existence of strong ionic attractions in the crystal lattice

II. The formation of strong ion-dipole attractions

III. An increase in entropy upon dissolving

(A) I only
(B) I and II only
(C) I and III only
(D) II and III only ✓
(E) I, II, and III

**26.** How many moles of NaF must be dissolved in 1.00 liter of a saturated solution of $PbF_2$ at 25°C to reduce the $[Pb^{2+}]$ to 1 x $10^{-6}$ molar?

($K_{sp}$ $PbF_2$ at 25°C = 4.0 x $10^{-8}$)

(A) 0.020 mole
(B) 0.040 mole
(C) 0.10 mole
(D) 0.20 mole ✓
(E) 0.40 mole

**27.** In a saturated solution of $Zn(OH)_2$ at 25°C, the value of $[OH^-]$ is 2.0 x $10^{-6}$ *M*. What is the value of the solubility-product constant, $K_{sp}$, for $Zn(OH)_2$ at 25°C ?

(A) 4.0 x $10^{-18}$ ✓
(B) 8.0 x $10^{-18}$
(C) 1.6 x $10^{-17}$
(D) 4.0 x $10^{-12}$
(E) 2.0 x $10^{-6}$

**28.** The value of $K_{sp}$ for the salt $PbF_2$ is $3.2 \times 10^{-8}$. The $[Pb^{2+}]$ in a saturated solution of $PbF_2$ is approximately

## Chapter 10 Part 3: Ksp

(A) $2 \times 10^{-4}\ M$

(B) $4 \times 10^{-4}\ M$

(C) $2 \times 10^{-3}\ M$

(D) $4 \times 10^{-3}\ M$

## Chapter 10 Part 3: Ksp

**29.**

| | |
|---|---|
| Mass of KI tablet | 0.425 g |
| Mass of thoroughly dried filter paper | 1.462 g |
| Mass of filter paper + precipitate after first drying | 1.775 g |
| Mass of filter paper + precipitate after second drying | 1.699 g |
| Mass of filter paper + precipitate after third drying | 1.698 g |

A student is given the task of determining the $I^-$ content of tablets that contain KI and an inert, water-soluble sugar as a filler. A tablet is dissolved in 50.0 mL of distilled water, and an excess of 0.20 $M$ $Pb(NO_3)_2(aq)$ is added to the solution. A yellow precipitate forms, which is then filtered, washed, and dried. The data from the experiment are shown in the table above.

a. For the chemical reaction that occurs when the precipitate forms,

   i. write a balanced, net-ionic equation for the reaction, and

   ii. explain why the reaction is best represented by a net-ionic equation.

b. Explain the purpose of drying and weighing the filter paper with the precipitate three times.

c. In the filtrate solution, is $[K^+]$ greater than, less than, or equal to $[NO_3^-]$ ? Justify your answer.

d. Calculate the number of moles of precipitate that is produced in the experiment.

e. Calculate the mass percent of $I^-$ in the tablet.

f. In another trial, the student dissolves a tablet in 55.0 mL of water instead of 50.0 mL of water. Predict whether the experimentally determined mass percent of $I^-$ will be greater than, less than, or equal to the amount calculated in part (e). Justify your answer.

g. A student in another lab also wants to determine the $I^-$ content of a KI tablet but does not have access to $Pb(NO_3)_2$ . However, the student does have access to 0.20 $M$ $AgNO_3$ , which reacts with $I^-$ $(aq)$ to produce $AgI(s)$. The value of $K_{sp}$ for AgI is $8.5 \times 10^{-17}$.

   i. Will the substitution of $AgNO_3$ for $Pb(NO_3)_2$ result in the precipitation of the $I^-$ ion from solution? Justify your answer.

   ii. The student only has access to one KI tablet and a balance that can measure to the nearest 0.01 g. Will the student be able to determine the mass of AgI produced to three significant figures? Justify your answer.

**Part A**

- 1 point is earned for a balanced net-ionic equation.

## Chapter 10 Part 3: Ksp

$Pb^{2+} + 2\,I^- \rightarrow PbI_2$

- 1 point is earned for a valid explanation.

The net-ionic equation shows the formation of the $PbI_2$(*s*) from $Pb^{2+}$(*aq*) and $I^-$(*aq*) ions, omitting the non-reacting species (spectator ions), $K^+$(*aq*) and $NO_3^-$(*aq*).

| 0 | 1 | 2 |
|---|---|---|

The student response earns two of the following points:

- 1 point is earned for a balanced net-ionic equation.

$Pb^{2+} + 2\,I^- \rightarrow PbI_2$

- 1 point is earned for a valid explanation.

The net-ionic equation shows the formation of the $PbI_2$(*s*) from $Pb^{2+}$(*aq*) and $I^-$(*aq*) ions, omitting the non-reacting species (spectator ions), $K^+$(*aq*) and $NO_3^-$(*aq*).

**Part B**

1 point is earned for a valid explanation.

The filter paper and precipitate must be dried several times (to a constant mass) to ensure that all the water has been driven off.

| 0 | 1 |
|---|---|

The student response earns one of the following points:

1 point is earned for a valid explanation.

The filter paper and precipitate must be dried several times (to a constant mass) to ensure that all the water has been driven off.

**Part C**

1 point is earned for a correct comparison with a valid explanation.

[$K^+$] is less than [$NO_3^-$] because the source of the $NO_3^-$, the 0.20 *M* $Pb(NO_3)_2$(*aq*), was added in excess.

## Chapter 10 Part 3: Ksp

| 0 | 1 |
|---|---|

The student response earns one of the following points:

1 point is earned for a correct comparison with a valid explanation.

[K+] is less than [NO3-] because the source of the NO3-, the 0.20 *M* Pb(NO3)2(*aq*), was added in excess.

**Part D**

1 point is earned for the correct number of moles of PbI2(*s*) precipitate.

1.698 g – 1.462 g = 0.236 g PbI2(*s*)

$$0.236 \text{ g PbI}_2 \times \frac{1 \text{ mol PbI}_2}{461.0 \text{ g PbI}_2} = 5.12 \times 10^{-4} \text{ mol PbI}_2$$

| 0 | 1 |
|---|---|

The student response earns one of the following points:

1 point is earned for the correct number of moles of PbI2(*s*) precipitate.

1.698 g – 1.462 g = 0.236 g PbI2(*s*)

$$0.236 \text{ g PbI}_2 \times \frac{1 \text{ mol PbI}_2}{461.0 \text{ g PbI}_2} = 5.12 \times 10^{-4} \text{ mol PbI}_2$$

**Part E**

1 point is earned for determining the number of moles of I- in one tablet.

$$5.12 \times 10^{-4} \text{mol PbI}_2 \times \frac{2 \text{ mol I}^-}{1 \text{ mol PbI}_2} = 1.02 \times 10^{-3} \text{ mol I}^-$$

$$1.02 \times 10^{-3} \text{ mol I}^- \times \frac{126.91 \text{ g I}^-}{1 \text{ mol I}^-} = 0.130 \text{ g I}^- \text{ in one tablet}$$

1 point is earned for calculating the mass percent of I- in the KI tablet.

$$\frac{0.130 \text{ g I}^-}{0.425 \text{ g K tablet}} = 0.306 = 30.6\% \text{ I}^- \text{ per KI tablet}$$

**Chapter 10 Part 3: Ksp**

| 0 | 1 | 2 |
|---|---|---|

The student response earns two of the following points:

1 point is earned for determining the number of moles of I- in one tablet.

$$5.12 \times 10^{-4} \text{mol PbI}_2 \times \frac{2 \text{ mol I}^-}{1 \text{ mol PbI}_2} = 1.02 \times 10^{-3} \text{ mol I}^-$$

$$1.02 \times 10^{-3} \text{ mol I}^- \times \frac{126.91 \text{ g I}^-}{1 \text{ mol I}^-} = 0.130 \text{ g I}^- \text{ in one tablet}$$

1 point is earned for calculating the mass percent of I- in the KI tablet.

$$\frac{0.130 \text{ g I}^-}{0.425 \text{ g K tablet}} = 0.306 = 30.6\,\% \text{ I}^- \text{ per KI tablet}$$

**Part F**

1 point is earned for correct comparison with a valid justification.

The mass percent of I- will be the same. Pb2+(*aq*) was added in excess, ensuring that essentially no I- remained in solution. The additional water is removed by filtration and drying, leaving the same mass of dried precipitate.

| 0 | 1 |
|---|---|

The student response earns one of the following points:

1 point is earned for correct comparison with a valid justification.

The mass percent of I- will be the same. Pb2+(*aq*) was added in excess, ensuring that essentially no I- remained in solution. The additional water is removed by filtration and drying, leaving the same mass of dried precipitate.

**Part G**

- 1 point is earned for the correct answer with a valid justification.

  Yes. Addition of an excess of 0.20 *M* AgNO3(*aq*) will precipitate all of the I- ion present in the solution because AgI is insoluble, as evidenced by its low value of *Ksp*.

- 1 point is earned for a correct answer with a valid justification.

  No. If masses can be measured to ±0.01 g, then the mass of the dry AgI(*s*) precipitate (which is less than 1 g) will be known to only two significant figures.

## Chapter 10 Part 3: Ksp

| 0 | 1 | 2 |
|---|---|---|

The student response earns two of the following points:

- 1 point is earned for the correct answer with a valid justification.

  Yes. Addition of an excess of 0.20 *M* AgNO3(*aq*) will precipitate all of the I- ion present in the solution because AgI is insoluble, as evidenced by its low value of *Ksp*.

- 1 point is earned for a correct answer with a valid justification.

  No. If masses can be measured to ±0.01 g, then the mass of the dry AgI(*s*) precipitate (which is less than 1 g) will be known to only two significant figures.

## Chapter 10 Part 3: Ksp

**30.**

| | |
|---|---|
| Mass of KI tablet | 0.425 g |
| Mass of thoroughly dried filter paper | 1.462 g |
| Mass of filter paper + precipitate after first drying | 1.775 g |
| Mass of filter paper + precipitate after second drying | 1.699 g |
| Mass of filter paper + precipitate after third drying | 1.698 g |

A student is given the task of determining the $I^-$ content of tablets that contain KI and an inert, water-soluble sugar as a filler. A tablet is dissolved in 50.0 mL of distilled water, and an excess of 0.20 *M* $Pb(NO_3)_2(aq)$ is added to the solution. A yellow precipitate forms, which is then filtered, washed, and dried. The data from the experiment are shown in the table above.

a. For the chemical reaction that occurs when the precipitate forms,

  i. write a balanced, net-ionic equation for the reaction, and

  ii. explain why the reaction is best represented by a net-ionic equation.

b. Explain the purpose of drying and weighing the filter paper with the precipitate three times.

c. In the filtrate solution, is $[K^+]$ greater than, less than, or equal to $[NO_3^-]$ ? Justify your answer.

d. Calculate the number of moles of precipitate that is produced in the experiment.

e. Calculate the mass percent of $I^-$ in the tablet.

f. In another trial, the student dissolves a tablet in 55.0 mL of water instead of 50.0 mL of water. Predict whether the experimentally determined mass percent of $I^-$ will be greater than, less than, or equal to the amount calculated in part (e). Justify your answer.

g. A student in another lab also wants to determine the $I^-$ content of a KI tablet but does not have access to $Pb(NO_3)_2$ . However, the student does have access to 0.20 *M* $AgNO_3$ , which reacts with $I^-(aq)$ to produce $AgI(s)$. The value of $K_{sp}$ for AgI is $8.5 \times 10^{-17}$.

  i. Will the substitution of $AgNO_3$ for $Pb(NO_3)_2$ result in the precipitation of the $I^-$ ion from solution? Justify your answer.

  ii. The student only has access to one KI tablet and a balance that can measure to the nearest 0.01 g. Will the student be able to determine the mass of AgI produced to three significant figures?

# Chapter 10 Part 3: Ksp

Justify your answer.

**Part A**

1 point is earned for a balanced net-ionic equation.

Pb2+ + 2 I− → PbI2

1 point is earned for a valid explanation.

The net-ionic equation shows the formation of the PbI2(*s*) from Pb2+(*aq*) and I−(*aq*) ions, omitting the non-reacting species (spectator ions), K+ (*aq*) and NO3–(*aq*).

| 0 | 1 | 2 |
|---|---|---|

The student response earns two of the following points:

1 point is earned for a balanced net-ionic equation.

Pb2+ + 2 I− → PbI2

1 point is earned for a valid explanation.

The net-ionic equation shows the formation of the PbI2(*s*) from Pb2+(*aq*) and I−(*aq*) ions, omitting the non-reacting species (spectator ions), K+ (*aq*) and NO3–(*aq*).

**Part B**

1 point is earned for a valid explanation.

The filter paper and precipitate must be dried several times (to a constant mass) to ensure that all the water has been driven off.

| 0 | 1 |
|---|---|

The student response earns one of the following points:

1 point is earned for a valid explanation.

The filter paper and precipitate must be dried several times (to a constant mass) to ensure that all the water has been driven off.

**Part C**

## Chapter 10 Part 3: Ksp

1 point is earned for a correct comparison with a valid explanation.

[K+] is less than [NO3−] because the source of the NO3−, the 0.20 *M* Pb(NO3)2(*aq*), was added in excess.

| 0 | 1 |
|---|---|

The student response earns one of the following points:

1 point is earned for a correct comparison with a valid explanation.

[K+] is less than [NO3−] because the source of the NO3−, the 0.20 *M* Pb(NO3)2(*aq*), was added in excess.

**Part D**

1 point is earned for the correct number of moles of PbI2(*s*) precipitate.

1.698 g − 1.462 g = 0.236 g PbI2(*s*)

$$0.236 \text{ g PbI}_2 \times \frac{1 \text{ mol PbI}_2}{461.0 \text{ g PbI}_2} = 5.12 \times 10^{-4} \text{ mol PbI}_2$$

| 0 | 1 |
|---|---|

The student response earns one of the following points:

1 point is earned for the correct number of moles of PbI2(*s*) precipitate.

1.698 g − 1.462 g = 0.236 g PbI2(*s*)

$$0.236 \text{ g PbI}_2 \times \frac{1 \text{ mol PbI}_2}{461.0 \text{ g PbI}_2} = 5.12 \times 10^{-4} \text{ mol PbI}_2$$

**Part E**

1 point is earned for determining the number of moles of I− in one tablet.

$$5.12 \times 10^{-4} \text{ mol PbI}_2 \times \frac{2 \text{ mol I}^-}{1 \text{ mol PbI}_2} = 1.02 \times 10^{-3} \text{ mol I}^-$$

$$1.02 \times 10^{-3} \text{ mol I}^- \times \frac{126.91 \text{ g I}^-}{1 \text{ mol I}^-} = 0.130 \text{ g I}^- \text{ in one tablet}$$

1 point is earned for calculating the mass percent of I− in the KI tablet

$$\frac{0.130 \text{ g I}^-}{0.425 \text{ g KI tablet}} = 0.306 = 30.6\% \text{ I}^- \text{ per KI tablet}$$

## Chapter 10 Part 3: Ksp

| 0 | 1 | 2 |
|---|---|---|

The student response earns two of the following points:

1 point is earned for determining the number of moles of I− in one tablet.

$$5.12 \times 10^{-4} \text{ mol PbI}_2 \times \frac{2 \text{ mol I}^-}{1 \text{ mol PbI}_2} = 1.02 \times 10^{-3} \text{ mol I}^-$$

$$1.02 \times 10^{-3} \text{ mol I}^- \times \frac{126.91 \text{ g I}^-}{1 \text{ mol I}^-} = 0.130 \text{ g I}^- \text{ in one tablet}$$

1 point is earned for calculating the mass percent of I− in the KI tablet

$$\frac{0.130 \text{ g I}^-}{0.425 \text{ g KI tablet}} = 0.306 = 30.6\% \text{ I}^- \text{ per KI tablet}$$

**Part F**

1 point is earned for correct comparison with a valid justification.

The mass percent of I− will be the same. Pb2+(*aq*) was added in excess, ensuring that essentially no I− remained in solution. The additional water is removed by filtration and drying, leaving the same mass of dried precipitate

| 0 | 1 |
|---|---|

The student response earns one of the following points:

1 point is earned for correct comparison with a valid justification.

The mass percent of I− will be the same. Pb2+(*aq*) was added in excess, ensuring that essentially no I− remained in solution. The additional water is removed by filtration and drying, leaving the same mass of dried precipitate

**Part G**

1 point is earned for the correct answer with a valid justification.

Yes. Addition of an excess of 0.20 *M* AgNO3(*aq*) will precipitate all of the I− ion present in the solution because AgI is insoluble, as evidenced by its low value of *Ksp*.

1 point is earned for the correct answer with a valid justification.

No. If masses can be measured to ±0.01 g, then the mass of the dry AgI(*s*) precipitate (which is less than 1 g) will be known to only two significant figures.

## Chapter 10 Part 3: Ksp

| 0 | 1 | 2 |
|---|---|---|

The student response earns two of the following points:

1 point is earned for the correct answer with a valid justification.

Yes. Addition of an excess of 0.20 *M* AgNO3(*aq*) will precipitate all of the I− ion present in the solution because AgI is insoluble, as evidenced by its low value of *Ksp*.

1 point is earned for the correct answer with a valid justification.

No. If masses can be measured to ±0.01 g, then the mass of the dry AgI(*s*) precipitate (which is less than 1 g) will be known to only two significant figures.

**31.** The value of $K_{sp}$ for $PbCl_2$ is $1.6 \times 10^{-5}$. What is the lowest concentration of $Cl^-(aq)$ that would be needed to begin precipitation of $PbCl_2(s)$ in 0.010 *M* $Pb(NO_3)_2$ ?

(A) $1.6 \times 10^{-7}$ *M*

(B) $4.0 \times 10^{-4}$ *M*

(C) $1.6 \times 10^{-3}$ *M*

(D) $2.6 \times 10^{-3}$ *M*

(E) $4.0 \times 10^{-2}$ *M*

**32.**

| Compound | $[Pb^{2+}]$ in Saturated Solution |
|---|---|
| PbS | $1.7 \times 10^{-14}\ M$ |
| $PbCO_3$ | $2.7 \times 10^{-7}\ M$ |
| $PbCrO_4$ | $5.5 \times 10^{-7}\ M$ |
| $PbSO_4$ | $1.6 \times 10^{-4}\ M$ |

Based on $[Pb^{2+}]$ in saturated solutions of the compounds listed in the table above, which of the compounds has the smallest $K_{sp}$ value?

(A) PbS

(B) $PbCO_3$

(C) $PbCrO_4$

(D) $PbSO_4$

## Chapter 10 Part 3: Ksp

**33.** $\mathrm{AgCl}(s) \rightleftarrows \mathrm{Ag}^{+}(aq) + \mathrm{Cl}^{-}(aq) \quad K_{sp} = 1.8 \times 10^{-10}$

Shown above is information about the dissolution of $\mathrm{AgCl}(s)$ in water at $298\ \mathrm{K}$. In a chemistry lab a student wants to determine the value of $s$, the molar solubility of $\mathrm{AgCl}$, by measuring $[\mathrm{Ag}^{+}]$ in a saturated solution prepared by mixing excess $\mathrm{AgCl}$ and distilled water. How would the results of the experiment be altered if the student mixed excess $\mathrm{AgCl}$ with tap water (in which $[\mathrm{Cl}^{-}] = 0.010\ M$) instead of distilled water and the student did not account for the $\mathrm{Cl}^{-}$ in the tap water?

(A) The value obtained for $K_{sp}$ would be too small because $\mathrm{Cl}^{-}(aq)$ ions would be attracted to the $\mathrm{Ag}^{+}$ ions in the $\mathrm{AgCl}$ crystals, thus preventing water molecules from reaching the crystals.

(B) The value obtained for $K_{sp}$ would be too small because less $\mathrm{AgCl}(s)$ would dissolve because of the common ion effect due to the $\mathrm{Cl}^{-}(aq)$ already in the water. ✓

(C) The value obtained for $K_{sp}$ would be too large because more $\mathrm{AgCl}(s)$ would dissolve because of the attractions between $\mathrm{Ag}^{+}$ ions in the $\mathrm{AgCl}$ crystals and the $\mathrm{Cl}^{-}(aq)$ ions in the water.

(D) The results of the experiment would not be altered because $0.010\ M$ is such a small concentration of $\mathrm{Cl}^{-}(aq)$ ions and thus has no effect on the dissolution of $\mathrm{AgCl}(s)$.

**34.**

| **Substance** | $K_{sp}$ |
|---|---|
| $\mathrm{Sn(OH)_2}$ | $K_{sp} = 4S^3 = 5.45 \times 10^{-27}$ |
| $\mathrm{CuCN}$ | $K_{sp} = S^2 = 3.47 \times 10^{-20}$ |
| $\mathrm{MgF_2}$ | $K_{sp} = 4S^3 = 5.16 \times 10^{-11}$ |
| $\mathrm{NiCO_3}$ | $K_{sp} = S^2 = 1.42 \times 10^{-7}$ |

The equilibrium constants for the dissolution ($K_{sp}$) of various substances in aqueous solution at $25^{\circ}\mathrm{C}$ are listed in the table above. Which of the following provides a correct comparison of the molar solubilities ($S$) of some of these substances based on their $K_{sp}$ ?

(A) The molar solubilities for $\mathrm{CuCN}$ and $\mathrm{NiCO_3}$ are calculated using $S = \sqrt{K_{sp}}$ and $\mathrm{NiCO_3}$ has a lower molar solubility than $\mathrm{CuCN}$.

(B) The molar solubilities for $\mathrm{CuCN}$ and $\mathrm{NiCO_3}$ are calculated using $S = \sqrt{K_{sp}}$ and $\mathrm{CuCN}$ has a lower molar solubility than $\mathrm{NiCO_3}$. ✓

(C) The molar solubilities for $\mathrm{Sn(OH)_2}$ and $\mathrm{MgF_2}$ are calculated using $S = \sqrt{\frac{K_{sp}}{4}}$ and $\mathrm{MgF_2}$ has a lower molar solubility than $\mathrm{NiCO_3}$.

(D) The molar solubilities for $\mathrm{Sn(OH)_2}$ and $\mathrm{MgF_2}$ are calculated using $S = \sqrt{\frac{K_{sp}}{4}}$ and $\mathrm{Sn(OH)_2}$ has a lower molar solubility than $\mathrm{NiCO_3}$.

# Chapter 10 Part 3: Ksp

**35.** $CaF_2(s) \rightleftarrows Ca^{2+}(aq) + 2\,F^{-}(aq) \qquad K_{sp} = 1.5 \times 10^{-11}$

The concentration of $F^{-}(aq)$ in drinking water that is considered to be ideal for promoting dental health is $4.0 \times 10^{-5}\ M$. Based on the information above, the maximum concentration of $Ca^{2+}(aq)$ that can be present in drinking water without lowering the concentration of $F^{-}(aq)$ below the ideal level is closest to

(A) $0.0023\ M$

(B) $0.0094\ M$ ✓

(C) $3.8 \times 10^{-7}\ M$

(D) $6.0 \times 10^{-16}\ M$

## Chapter 10 Part 3: Ksp

**36. Directions: For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.**

The majority of elements in the periodic table are metals. At $298\,\mathrm{K}$ most metals are solid and share similar physical properties.

(a) A student claims that gold is a good conductor of electricity because it is dense. Do you agree or disagree with this claim? Justify your answer.

(b) In a photoelectron spectroscopy experiment, photons with wavelength $130\,\mathrm{nm}$ cause the ejection of $6\,s$ electrons from gold, but photons with wavelength $105\,\mathrm{nm}$ cause the electrons to be ejected with greater energy. A student claims this is because the photons with wavelength $130\,\mathrm{nm}$ are higher in energy than photons with wavelength $105\,\mathrm{nm}$. Do you agree or disagree with the student's claim? Justify your answer in terms of the energies of the photons.

A student investigates a pure metal, $\mathrm{X}$. The student takes a $100.0\,\mathrm{g}$ piece of metal $\mathrm{X}$, heats it to $500.0\,^{\circ}\mathrm{C}$, then places it on a $1000.0\,\mathrm{g}$ block of ice at $0.0\,^{\circ}\mathrm{C}$. The ice partially melts, and the final temperature of the metal, ice, and melted water is $0.0\,^{\circ}\mathrm{C}$. The student calculates the experimental value of the specific heat capacity of metal $\mathrm{X}$ and records it as $0.13\,\mathrm{J}/(\mathrm{g}\cdot\,^{\circ}\mathrm{C})$.

(c) Calculate the magnitude of the energy change of metal $\mathrm{X}$ during the experiment.

(d) Calculate the mass of ice melted by metal $\mathrm{X}$. (The enthalpy of fusion of ice is $330\,\mathrm{J}/\mathrm{g}$.)

(e) A student claims that the energy transferred to the ice is used to overcome hydrogen bonding between water molecules in the solid ice. Do you agree or disagree with this claim? Justify your answer.

The student takes a sample of $\mathrm{XCl_2}$, the chloride salt of metal $\mathrm{X}$, and mixes it with $100.0\,\mathrm{mL}$ of distilled water to form a saturated solution. The value of $K_{sp}$ for $\mathrm{XCl_2}$ is $4.0\times10^{-12}$ at $298\,\mathrm{K}$.

(f) Write the balanced chemical equation for the change that occurs when solid $\mathrm{XCl_2}$ dissolves in water.

(g) Calculate the concentration of $\mathrm{Cl^-}(aq)$ ions in the saturated solution.

(h) Will the solubility of $\mathrm{XCl_2}$ increase, decrease, or remain the same if it is dissolved in $100.0\,\mathrm{mL}$ of $0.10\,M\,\mathrm{NaCl}$ instead of distilled water? Explain.

(i) The student gently heats $100.0\,\mathrm{mL}$ of the saturated solution of $\mathrm{XCl_2}$ to a temperature of $315\,\mathrm{K}$. If the enthalpy change of the dissolution process is positive, will the concentration of $\mathrm{X^{2+}}(aq)$ ions in the solution be greater than, less than, or the same as when the temperature of the solution was $298\,\mathrm{K}$? Justify your answer.

The first five ionization energies for metal $\mathrm{X}$ are shown in the following table.

**Chapter 10 Part 3: Ksp**

| Ionization Energies of Metal $X$ $(kJ/mol)$ | | | | |
|---|---|---|---|---|
| First | Second | Third | Fourth | Fifth |
| 590 | 1100 | 4900 | 6500 | 8200 |

(j) The ionic compound $XCl_3$ is not known to exist. Explain why, using the data in the table above to support your explanation.

For a compound, the energy required to separate the ions in a crystal lattice into individual gaseous ions is known as the lattice energy of the compound. The following table shows the dissociation reactions and the lattice energies for $XF_2$ and for $ZF_2$. ($Z$ is another metal different from metal $X$.)

| Dissociation Reaction | Lattice Energy $(kJ/mol)$ |
|---|---|
| $XF_2(s) \rightarrow X^{2+}(g) + 2\,F^-(g)$ | 2609 |
| $ZF_2(s) \rightarrow Z^{2+}(g) + 2\,F^-(g)$ | 2341 |

(k) Using the information in the table above, predict whether the ionic radius of $X^{2+}$ is greater than, less than, or equal to the ionic radius of $Z^{2+}$. Justify your prediction in terms of periodic properties and Coulomb's law.

**Part A**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides both the correct choice and a valid justification equivalent to the following.

· Disagree. The density of gold is not the reason that it is a good electrical conductor. Electrical conductivity is related to the mobility of the loosely held valence electrons that surround the positive metal cores like a "sea of electrons."

**Part B**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides both the correct choice and a valid justification equivalent to the following.

· Disagree. The photons with wavelength $130\ nm$ have a lower frequency than those with wavelength $105\ nm$ because wavelength and frequency are inversely proportional $(\nu = \frac{c}{\lambda})$. Since frequency and photon energy are directly proportional $(E = h\,\nu)$, the photons with wavelength $130\ \mathrm{nm}$ are lower in energy than those with wavelength of $105\ \mathrm{nm}$.

**Part C**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides a correct calculation of the magnitude of $q$.

· $q_{\text{lost by metal}} = mc\Delta T = (100.0\,\mathrm{g})(0.13\,\frac{\mathrm{J}}{\mathrm{g}\cdot{}^{\circ}\mathrm{C}})(500.0\,{}^{\circ}\mathrm{C})$

$q_{\text{lost by metal}} = 6500\,\mathrm{J}$

**Part D**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides a correct calculation of mass, consistent with the answer in part (c).

· $|q_{\text{lost by metal}}| = |q_{\text{gained by ice}}|$

$6500\,\mathrm{J} = (m_{\text{ice}})(\Delta H_{fusion}) = (m_{\text{ice}})(330\,\mathrm{J/g})$

$m_{\text{ice}} = 20.\,\mathrm{g}$

**Part E**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides both the correct choice and a valid justification equivalent to the following.

· Agree. Intermolecular hydrogen bonds are primarily responsible for holding the water molecules in regular arrangement in solid ice, and some hydrogen bonds are overcome as ice transitions to the liquid phase.

**Part F**

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Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides the correct balanced equation (state symbols are not required).

· $XCl_2(s) \rightleftarrows X^{2+}(aq) + 2\,Cl^-(aq)$

**Part G**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 | 2 |
|---|---|---|

The response meets **both** of the following criteria.

- The response identifies the relationship between $K_{sp}$ and molar solubility.
  · $K_{sp} = [X^{2+}][Cl^-]^2$
  $4.0 \times 10^{-12} = (x)(2x)^2 = 4x^3$
- The response shows the correct calculation of the concentration of $Cl^-$ ions.
  · $4.0 \times 10^{-12} = 4x^3$
  $1.0 \times 10^{-12} = x^3$
  $1.0 \times 10^{-4} M = x$
  $[Cl^-] = 2x = 2(1.0 \times 10^{-4} M) = 2.0 \times 10^{-4} M$

**Part H**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides both the correct answer and a valid explanation equivalent to one of the following.

· The solubility of $XCl_2$ will decrease because of the presence of the common ion, $Cl^-(aq)$, in solution. The presence of the common ion favors a shift towards the reactants, reducing the dissolution of $XCl_2$.

OR

· The solubility of $XCl_2$ will decrease because of the presence of $Cl^-(aq)$ ions in the solution. Since $K_{sp}$ is constant

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and $K_{sp} = [\mathrm{X}^{2+}][\mathrm{Cl}^{-}]^{2}$, larger values of $[\mathrm{Cl}^{-}]$ correspond to smaller values of $[\mathrm{X}^{2+}]$.

**Part I**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides both the correct choice and a valid justification.

· $[\mathrm{X}^{2+}]$ will be greater at $315\ \mathrm{K}$. According to Le Châtelier's principle, the distribution of species will shift towards products when the temperature is increased in an endothermic process.

**Part J**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides a correct explanation equivalent to the following.

· The energies required to remove the first and second electrons from an atom of metal X are much less than the energy required to remove the third electron, making a $+3$ ionic charge in a compound like $\mathrm{XCl_3}$ very unlikely.

**Part K**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response gives a justification that links stronger coulombic attractions to smaller cations equivalent to the following.

· The ionic radius of $\mathrm{X}^{2+}$ must be less than that of $\mathrm{Z}^{2+}$. Because of its smaller radius, the $\mathrm{X}^{2+}$ ions are closer to the $\mathrm{F}^{-}$ ions in the crystal lattice. By Coulomb's law, a shorter interionic distance implies stronger interionic attractions. Therefore, more energy is required to separate the ions, and $\mathrm{XF_2}$ has the higher lattice energy.

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**37. Directions: For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.**

The majority of elements in the periodic table are metals. At $298\ \mathrm{K}$ most metals are solid and share similar physical properties.

(a) A student claims that gold is malleable because it is dense. Do you agree or disagree with this claim? Justify your answer.

(b) In a photoelectron spectroscopy experiment, photons with wavelength $130\ \mathrm{nm}$ cause the ejection of $6\ s$ electrons from gold, but photons with wavelength $200\ \mathrm{nm}$ do not. A student claims this is because the photons with wavelength $200\ \mathrm{nm}$ are lower in energy than photons with wavelength $130\ \mathrm{nm}$. Do you agree or disagree with the student's claim? Justify your answer in terms of the energies of the photons.

A student investigates a pure metal, $\mathrm{X}$. The student takes a $100.0\ \mathrm{g}$ piece of metal $\mathrm{X}$, heats it to $500.0^{\circ}\mathrm{C}$, then places it on a $1000.0\ \mathrm{g}$ block of ice at $0.0^{\circ}\mathrm{C}$. The ice partially melts, and the final temperature of the metal, ice, and melted water is $0.0^{\circ}\mathrm{C}$. The student calculates the experimental value of the specific heat capacity of metal $\mathrm{X}$ and records it as $0.90\ \mathrm{J/(g \cdot {}^{\circ}C)}$.

(c) Calculate the magnitude of the energy change of metal $\mathrm{X}$ during the experiment.

(d) Calculate the mass of ice melted by metal $\mathrm{X}$. (The enthalpy of fusion of ice is $330\ \mathrm{J/g}$.)

(e) A student claims that the energy transferred to the ice breaks the polar-covalent $\mathrm{O-H}$ bonds found between atoms within water molecules. Do you agree or disagree with this claim? Justify your answer.

The student takes a sample of $\mathrm{XF_2}$, the fluoride salt of metal $\mathrm{X}$, and mixes it with $100.0\ \mathrm{mL}$ of distilled water to form a saturated solution. The value of $K_{sp}$ for $\mathrm{XF_2}$ is $4.0 \times 10^{-9}$ at $298\ \mathrm{K}$.

(f) Write the balanced chemical equation for the change that occurs when solid $\mathrm{XF_2}$ dissolves in water.

(g) Calculate the concentration of $\mathrm{F^-}(aq)$ ions in the saturated solution.

(h) Will the solubility of $\mathrm{XF_2}$ increase, decrease, or remain the same if it is dissolved in $100.0\ \mathrm{mL}$ of $0.10\ M\ \mathrm{NaF}$ instead of distilled water? Explain.

(i) The student gently heats $100.0\ \mathrm{mL}$ of the saturated solution of $\mathrm{XF_2}$, evaporating enough water to reduce the volume of the solution to $75.0\ \mathrm{mL}$. After the temperature of the solution returns to $298\ \mathrm{K}$, will the concentration of $\mathrm{X^{2+}}(aq)$ ions in the solution be greater than, less than, or the same as when the volume of the solution was $100.0\ \mathrm{mL}$? Justify your answer.

The first five ionization energies for metal $\mathrm{X}$ are shown in the following table.

## Chapter 10 Part 3: Ksp

| Ionization Energies of Metal X (kJ/mol) | | | | |
|---|---|---|---|---|
| First | Second | Third | Fourth | Fifth |
| 736 | 1445 | 7730 | 10,500 | 13,600 |

(j) The ionic compound $XF_3$ is not known to exist. Explain why, using the data in the table above to support your explanation.

For a compound, the energy required to separate the ions in a crystal lattice into individual gaseous ions is known as the lattice energy of the compound. The following table shows the dissociation reactions and the lattice energies for $XCl_2$ and for $XBr_2$.

| Dissociation Reaction | Lattice Energy (kJ/mol) |
|---|---|
| $XCl_2(s) \rightarrow X^{2+}(g) + 2\,Cl^-(g)$ | 2524 |
| $XBr_2(s) \rightarrow X^{2+}(g) + 2\,Br^-(g)$ | 2440 |

(k) Explain in terms of atomic properties and Coulomb's law why the lattice energy of $XCl_2$ is greater than that of $XBr_2$.

**Part A**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides both the correct choice and a valid justification equivalent to the following.

· Disagree. Malleability is a result of the delocalized electrons in metals, not the density. Because valence electrons in gold are delocalized, the remaining positive metal cores can move past one another without a significant change in the structure.

**Part B**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides both the correct choice and a valid justification equivalent to the following.

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· Agree. The photons with wavelength $200\ \mathrm{nm}$ have a lower frequency than those with wavelength $130\ \mathrm{nm}$ because frequency and wavelength are inversely proportional $(\nu = \frac{c}{\lambda})$. Since frequency and photon energy are directly proportional $(E = h\,\nu)$, the lower frequency photons with wavelength $200\ \mathrm{nm}$ are lower in energy than those with wavelength $130\ \mathrm{nm}$ .

**Part C**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides a correct calculation of the magnitude of $q$.

· $q_{\text{lost by metal}} = mc\Delta T = (100.0\,\mathrm{g})(0.90\,\frac{\mathrm{J}}{\mathrm{g}\cdot{}^\circ\mathrm{C}})(500.0\,{}^\circ\mathrm{C})$

$q_{\text{lost by metal}} = 45,000\,\mathrm{J}$

**Part D**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides a correct calculation of mass, consistent with the answer in part (c).

· $|q_{\text{lost by metal}}| = |q_{\text{gained by ice}}|$

$45,000\,\mathrm{J} = (m_{\text{ice}})(\Delta H_{fusion}) = (m_{\text{ice}})(330\,\mathrm{J/g})$

$m_{\text{ice}} = 140\,\mathrm{g}$

**Part E**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides both the correct choice and a valid justification equivalent to the following.

· Disagree. The energy goes into overcoming intermolecular forces as ice is converted to liquid water. Chemical bonds are not broken during a phase change.

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### Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides the correct balanced equation (state symbols are not required).

· $XF_2(s) \rightleftarrows X^{2+}(aq) + 2\,F^-(aq)$

### Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 | 2 |
|---|---|---|

The response meets **both** of the following criteria.

- The response identifies the relationship between $K_{sp}$ and molar solubility.
  · $K_{sp} = [X^{2+}][F^-]^2$
  $4.0 \times 10^{-9} = (x)(2x)^2 = 4x^3$
- The response shows the correct calculation of the concentration of $F^-$ ions.
  · $4.0 \times 10^{-9} = 4x^3$
  $1.0 \times 10^{-9} = x^3$
  $1.0 \times 10^{-3} M = x$
  $[F^-] = 2x = 2(1.0 \times 10^{-3} M) = 2.0 \times 10^{-3} M$

### Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides both the correct answer and a valid explanation equivalent to one of the following.

· The solubility of $XF_2$ will decrease because of the presence of the common ion, $F^-(aq)$, in solution. The presence of the common ion favors a shift towards the reactants, reducing the dissolution of $XF_2$.

OR

## Chapter 10 Part 3: Ksp

· The solubility of $XF_2$ would decrease because of the presence of $F^-(aq)$ ions in the solution. Since $K_{sp}$ is constant and $K_{sp} = [X^{2+}][F^-]^2$, larger values of $[F^-]$ correspond to smaller values of $[X^{2+}]$.

**Part I**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides both the correct choice and a valid justification.

· $[X^{2+}]$ remains the same. When solvent is removed from the solution, $Q$ increases and now $Q > K_{sp}$, thus more solid $XF_2$ will precipitate out of solution until equilibrium is reestablished.

**Part J**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides a correct explanation equivalent to the following.

· The energies required to remove the first and second electrons from an atom of metal $X$ are much less than the energy required to remove the third electron, making a $+3$ ionic charge in a compound like $XF_3$ very unlikely.

**Part K**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides a justification that links stronger coulombic attractions to smaller anions equivalent to the following.

· $Cl^-$ ions have smaller radii than $Br^-$ ions because they have one fewer occupied electron shell. Therefore, the distance between the centers of ions in the $XCl_2$ crystal is less than that in the $XBr_2$ crystal. According to Coulomb's law, the energy required to separate ions is inversely proportional to the distance between them.

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**38. Directions: For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.**

The majority of elements in the periodic table are metals. At $298\ \mathrm{K}$ most metals are solid and share similar physical properties.

(a) A student claims that silver is a good conductor of electricity because the valence electrons are strongly attracted to the nucleus. Do you agree or disagree with this claim? Justify your answer.

(b) In a photoelectron spectroscopy experiment, photons with frequency $1.2 \times 10^{15}\ \mathrm{Hz}$ cause the ejection of $5\,s$ electrons from silver, but photons with frequency $1.0 \times 10^{15}\ \mathrm{Hz}$ do not. A student claims this is because the photons with frequency $1.0 \times 10^{15}\ \mathrm{Hz}$ are higher in energy than photons with frequency $1.2 \times 10^{15}\ \mathrm{Hz}$. Do you agree or disagree with the student's claim? Justify your answer in terms of the energies of the photons.

A student investigates a pure metal, $\mathrm{X}$. The student takes a $100.0\ \mathrm{g}$ piece of metal $\mathrm{X}$, heats it to $500.0^{\circ}\mathrm{C}$, then places it on a $1000.0\ \mathrm{g}$ block of ice at $0.0^{\circ}\mathrm{C}$. The ice partially melts, and the final temperature of the metal, ice, and melted water is $0.0^{\circ}\mathrm{C}$. The student calculates the experimental value of the specific heat capacity of metal $\mathrm{X}$ and records it as $0.44\ \mathrm{J/(g \cdot {}^{\circ}C)}$.

(c) Calculate the magnitude of the energy change of metal $\mathrm{X}$ during the experiment.

(d) Calculate the mass of ice melted by metal $\mathrm{X}$. (The enthalpy of fusion of ice is $330\ \mathrm{J/g}$.)

(e) A student claims that the energy transferred to the ice overcomes attractions between $\mathrm{H^{+}}(aq)$ ions and $\mathrm{OH^{-}}(aq)$ ions in the ice. Do you agree or disagree with this claim? Justify your answer.

The student takes a sample of $\mathrm{Z_2SO_4}$, a compound of a different metal, $\mathrm{Z}$, and mixes it with $100.0\ \mathrm{mL}$ of water to form a saturated solution. The value of $K_{sp}$ for $\mathrm{Z_2SO_4}$ is $4.0 \times 10^{-6}$ at $25^{\circ}\mathrm{C}$.

(f) Write the balanced chemical equation for the change that occurs when solid $\mathrm{Z_2SO_4}$ dissolves in water.

(g) Calculate the concentration of $\mathrm{SO_4{}^{2-}}(aq)$ ions in the saturated solution.

(h) Will the solubility of $\mathrm{Z_2SO_4}$ increase, decrease, or remain the same if it is dissolved in $100.0\ \mathrm{mL}$ of $0.10\ M\ \mathrm{Na_2SO_4}$ instead of distilled water? Explain.

(i) The student then cools the saturated solution gently to a temperature of $10^{\circ}\mathrm{C}$. If the enthalpy change of the dissolution process is negative, will the concentration of $\mathrm{Z^{+}}(aq)$ ions in the solution be greater than, less than, or the same as when the temperature of the solution was $25^{\circ}\mathrm{C}$? Justify your answer.

The first five ionization energies for metal $\mathrm{Z}$ are shown in the following table.

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| Ionization Energies of Metal $Z$ ($kJ/mol$) | | | | |
|---|---|---|---|---|
| First | Second | Third | Fourth | Fifth |
| 403 | 2633 | 3787 | 5037 | 6603 |

(j) The ionic compound $ZF_2$ is not known to exist. Explain why, using the data in the table above to support your explanation.

For a compound, the energy required to separate the ions in a crystal lattice into individual gaseous ions is known as the lattice energy of the compound. The following table shows the dissociation reactions and the lattice energies for $MCl_2$ and for $MCl_3$, which are chlorides of a different metal, $M$.

| Dissociation Reaction | Lattice Energy ($kJ/mol$) |
|---|---|
| $MCl_2(s) \rightarrow M^{2+}(g) + 2\,Cl^-(g)$ | 2631 |
| $MCl_3(s) \rightarrow M^{3+}(g) + 3\,Cl^-(g)$ | ? |

(k) A student predicts that the lattice energy of $MCl_3$ is greater than the lattice energy of $MCl_2$. Do you agree with the student's prediction? Explain your answer in terms of atomic properties and Coulomb's law.

**Part A**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides both the correct choice and a valid justification equivalent to the following.

· Disagree. The valence electrons in silver are loosely attracted to the nucleus (delocalized). The mobility of the loosely held valence electrons that surround the metal atoms like a "sea of electrons" allows the conduction of an electric current through the solid metal.

**Part B**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides both the correct choice and a valid justification equivalent to the following.

· Disagree. Because frequency and photon energy are directly proportional $(E = h\nu)$, the photons with frequency

## Chapter 10 Part 3: Ksp

$1.0 \times 10^{15}$Hz are lower in energy than the photons with frequency $1.2 \times 10^{15}$Hz.

**Part C**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides a correct calculation of the magnitude of $q$.

· $q_{\text{lost by metal}} = mc\Delta T = (100.0\,\text{g})(0.44\,\frac{\text{J}}{\text{g}\cdot{}^\circ\text{C}})(500.0\,{}^\circ\text{C})$

$q_{\text{lost by metal}} = 22,000\,\text{J}$

**Part D**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides the correct calculation of mass, consistent with the answer in part (c).

· $|q_{\text{lost by metal}}| = |q_{\text{gained by ice}}|$

$22,000\,\text{J} = (m_{\text{ice}})(\Delta H_{fusion}) = (m_{\text{ice}})(330\,\text{J/g})$
$m_{\text{ice}} = 67\,\text{g}$

**Part E**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides both the correct choice and a valid justification equivalent to the following.

· Disagree. In solid water, no significant amount of $H^+(aq)$ or $OH^-(aq)$ is present. Rather, as ice at $0\,{}^\circ C$ absorbs energy, the energy overcomes the intermolecular forces between water molecules that keep the molecules in a fixed arrangement.

**Part F**

## Chapter 10 Part 3: Ksp

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides the correct balanced equation (state symbols are not required).

· $Z_2SO_4(s) \rightleftarrows 2\,Z^+(aq) + SO_4{}^{2-}(aq)$

**Part G**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 | 2 |
|---|---|---|

The response meets **both** of the following criteria.

- The response identifies the relationship between $K_{sp}$ and molar solubility.
  · $K_{sp} = [Z^+]^2[SO_4{}^{2-}]$
  $4.0 \times 10^{-6} = (x)(2x)^2 = 4x^3$
- The response shows the correct calculation of the concentration of $SO_4{}^{2-}$ ions.
  · $4.0 \times 10^{-6} = 4x^3$
  $1.0 \times 10^{-6} = x^3$
  $1.0 \times 10^{-2} M = x = [SO_4{}^{2-}]$

**Part H**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides both the correct answer and a valid explanation equivalent to one of the following.

· The solubility of $Z_2SO_4$ will decrease because of the presence of the common ion, $SO_4{}^{2-}(aq)$, in solution. The presence of the common ion favors a shift towards the reactants, reducing the dissolution of $Z_2SO_4$.

OR

· The solubility of $Z_2SO_4$ would decrease because of the presence of $SO_4{}^{2-}(aq)$ ions in the solution. Since $K_{sp}$ is constant and $K_{sp} = [Z^+]^2[SO_4{}^{2-}]$, larger values of $[SO_4{}^{2-}]$ correspond to smaller values of $[Z^+]$.

# Chapter 10 Part 3: Ksp

## Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides both the correct choice and a valid justification.

· $[Z^+]$ will be greater at $10^\circ C$. According to Le Châtelier's principle, the distribution of species will shift towards products when the temperature is decreased in an exothermic process.

## Part J

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides a correct explanation equivalent to the following.

· The energy required to remove the first electron from an atom of metal Z is much less than the energy required to remove the second electron, making a $+2$ ionic charge in a compound like $ZF_2$ very unlikely.

## Part K

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response gives a justification that links stronger coulombic attractions to smaller cations or greater magnitude of ionic charge, equivalent to one of the following.

· Agree. The ionic radius of $M^{3+}$ is smaller than that of $M^{2+}$, so the ions in $MCl_3$ lattice are closer to one another than in the $MCl_2$ lattice. According to Coulomb's law, the energy required to separate ions is inversely proportional to the distance between them, so the lattice energy of $MCl_3$ should be greater than that of $MCl_2$.

OR

· Agree. The $M^{3+}$ ion has a greater charge than $M^{2+}$. According to Coulomb's law, greater ionic charge leads to stronger interionic attractions. Correspondingly, more energy is required to separate the ions in $MCl_3$ than in $MCl_2$, leading to a greater lattice energy for $MCl_3$.

## Chapter 10 Part 3: Ksp

**39. Directions: For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.**

The majority of elements in the periodic table are metals. At $298\ \mathrm{K}$ most metals are solid and share similar physical properties.

(a) A student claims that silver is malleable because the valence electrons are strongly attracted to the nucleus. Do you agree or disagree with this claim? Justify your answer.

(b) In a photoelectron spectroscopy experiment, photons with frequency $1.2 \times 10^{15}\ \mathrm{Hz}$ cause the ejection of $5\ s$ electrons from silver, but photons with frequency $1.9 \times 10^{15}\ \mathrm{Hz}$ cause the electrons to be ejected with greater energy. A student claims this is because the photons with frequency $1.9 \times 10^{15}\ \mathrm{Hz}$ are higher in energy than photons with frequency $1.2 \times 10^{15}\ \mathrm{Hz}$. Do you agree or disagree with the student's claim? Justify your answer in terms of the energies of the photons.

A student investigates a pure metal, $\mathrm{X}$. The student takes a $100.0\ \mathrm{g}$ piece of metal $\mathrm{X}$, heats it to $500.0\,^{\circ}\mathrm{C}$, then places it on a $1000.0\ \mathrm{g}$ block of ice at $0.0\,^{\circ}\mathrm{C}$. The ice partially melts, and the final temperature of the metal, ice, and melted water is $0.0\,^{\circ}\mathrm{C}$. The student calculates the experimental value of the specific heat capacity of metal $\mathrm{X}$ and records it as $0.24\ \mathrm{J}/(\mathrm{g}\cdot{}^{\circ}\mathrm{C})$.

(c) Calculate the magnitude of the energy change of metal $\mathrm{X}$ during the experiment.

(d) Calculate the mass of ice melted by metal $\mathrm{X}$. (The enthalpy of fusion of ice is $330\ \mathrm{J}/\mathrm{g}$.)

(e) A student claims that the energy transferred to the ice overcomes only London dispersion forces between the water molecules in the solid ice. Do you agree or disagree with this claim? Justify your answer.

The student takes a sample of $\mathrm{Z_2S}$, the sulfide of a different metal, $\mathrm{Z}$, and mixes it with $100.0\ \mathrm{mL}$ of water to form a saturated solution. The value of $K_{sp}$ for $\mathrm{Z_2S}$ is $4.0 \times 10^{-30}$ at $298\ \mathrm{K}$.

(f) Write the balanced chemical equation for the change that occurs when solid $\mathrm{Z_2S}$ dissolves in water.

(g) Calculate the concentration of $\mathrm{S}^{2-}(aq)$ ions in the saturated solution.

(h) Will the solubility of $\mathrm{Z_2S}$ increase, decrease, or remain the same if it is dissolved in $100.0\ \mathrm{mL}$ of $0.10\ M\ \mathrm{Na_2S}$ instead of distilled water? Explain.

(i) The student then gently heats $100.0\ \mathrm{mL}$ of the saturated solution to a temperature of $315\ \mathrm{K}$. If the enthalpy change of the dissolution process is negative, will the concentration of $\mathrm{Z}^{+}(aq)$ ions in the solution be greater than, less than, or the same as when the temperature of the solution was $298\ \mathrm{K}$? Justify your answer.

The first five ionization energies for metal $\mathrm{Z}$ are shown in the following table.

## Chapter 10 Part 3: Ksp

| Ionization Energies of Metal $Z$ $(kJ/mol)$ | | | | |
|---|---|---|---|---|
| First | Second | Third | Fourth | Fifth |
| 419 | 3052 | 4410 | 5900 | 8000 |

(j) The ionic compound $ZF_2$ is not known to exist. Explain why, using the data in the table above to support your explanation.

For a compound, the energy required to separate the ions in a crystal lattice into individual gaseous ions is known as the lattice energy of the compound. The following table shows the dissociation reactions and the lattice energies for $ZCl$ and for $Z_2O$.

| Dissociation Reaction | Lattice Energy $(kJ/mol)$ |
|---|---|
| $ZCl(s) \rightarrow Z^+(g) + Cl^-(g)$ | 829 |
| $Z_2O(s) \rightarrow 2\,Z^+(g) + O^{2-}(g)$ | ? |

(k) A student predicts that the lattice energy of $ZCl$ is greater than the lattice energy of $Z_2O$. If the ionic radii of $Cl^-$ and $O^{2-}$ are approximately the same, is the student's prediction correct? Explain your answer in terms of periodic properties and Coulomb's law.

### Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides both the correct choice and a valid justification equivalent to the following.

· Disagree. Silver's valence electrons are loosely held (delocalized). Because of their delocalized valence electrons, the interatomic bonding in metals is not in any specific direction. As the metal is hammered, its atoms can slide past one another while maintaining their bonding with neighboring atoms.

### Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides both the correct choice and a valid justification equivalent to the following.

## Chapter 10 Part 3: Ksp

· Agree. Because frequency and photon energy are directly proportional ($E = h\nu$), the photons with frequency $1.9 \times 10^{15}$ Hz are higher in energy than the photons with frequency $1.2 \times 10^{15}$ Hz.

**Part C**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 ✓ |
|---|---|

The response provides a correct calculation of the magnitude of $q$.

· $q_{\text{lost by metal}} = mc\Delta T = (100.0\,\text{g})(0.24\,\frac{\text{J}}{\text{g}\cdot{}^{\circ}\text{C}})(500.0\,{}^{\circ}\text{C})$

$q_{\text{lost by metal}} = 12{,}000$ J

**Part D**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 ✓ |
|---|---|

The response provides a correct calculation of mass, consistent with the answer in part (c).

· $|q_{\text{lost by metal}}| = |q_{\text{gained by ice}}|$

$12{,}000\,\text{J} = (m_{\text{ice}})(\Delta H_{fusion}) = (m_{\text{ice}})(330\,\text{J/g})$
$m_{\text{ice}} = 36\,\text{g}$

**Part E**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 ✓ |
|---|---|

The response provides both the correct choice and a valid justification equivalent to the following.

· Disagree. Although London dispersion forces do exist between water molecules, the much stronger intermolecular hydrogen bonds are primarily responsible for holding the water molecules together in ice.

**Part F**

## Chapter 10 Part 3: Ksp

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides the correct balanced equation (state symbols are not required).

· $Z_2S(s) \rightleftarrows 2\,Z^+(aq) + S^{2-}(aq)$

**Part G**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 | 2 |
|---|---|---|

The response meets **both** of the following criteria.

- The response identifies the relationship between $K_{sp}$ and molar solubility.
  · $K_{sp} = [Z^+]^2[S^{2-}]$
  $4 \times 10^{-30} = (x)(2x)^2 = 4x^3$
- The response shows the correct calculation of the concentration of $S^{2-}$ ions.
  · $4 \times 10^{-30} = 4x^3$
  $1 \times 10^{-30} = x^3$
  $[S^{2-}] = x = 1 \times 10^{-10}$

**Part H**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides both the correct answer and a valid explanation equivalent to one of the following.

· The solubility of $Z_2S$ will decrease because of the presence of the common ion, $S^{2-}(aq)$, in solution. The presence of the common ion favors a shift towards the reactants, reducing the dissolution of $Z_2S$.

OR

· The solubility of $Z_2S$ will decrease because of the presence of $S^{2-}(aq)$ ions in the solution. Since $K_{sp}$ is constant and $K_{sp} = [Z^+]^2[S^{2-}]$, larger values of $[S^{2-}]$ correspond to smaller values of $[Z^+]$.

## Chapter 10 Part 3: Ksp

### Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides both the correct choice and a valid justification.

· $[Z^+]$ will be less at $315\ K$. According to Le Châtelier's principle, the distribution of species will shift towards reactants when the temperature is increased in an exothermic process.

### Part J

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides a correct explanation equivalent to the following.

· The energy required to remove the first electron from an atom of metal Z is much less than the energy required to remove the second electron, making a $+2$ ionic charge in a compound like $ZF_2$ very unlikely.

### Part K

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response gives a justification that links weaker coulombic attractions to lower magnitude of ionic charge equivalent to the following.

· No. The magnitude of the ionic charge of the $Cl^-$ ion is less than that of the $O^{2-}$ ion. According to Coulomb's law, the attraction between $Z^{2+}$ and the anion will be weaker in $ZCl$ than in $Z_2O$. Therefore, less energy is required to separate the ions in $ZCl$, and $ZCl$ will have a <u>lower</u> lattice energy than $Z_2O$.

## Chapter 10 Part 3: Ksp

**40.**

$$Mg(OH)_2(s) \rightleftarrows Mg^{2+}(aq) + 2\ OH^-\ (aq)$$

The exothermic dissolution of $Mg(OH)_2(s)$ in water is represented by the equation above. The $K_{sp}$ of $Mg(OH)_2$ is $1.8 \times 10^{-11}$. Which of the following changes will increase the solubility of $Mg(OH)_2$ in an aqueous solution?

(A) Decreasing the pH ✓

(B) Increasing the pH

(C) Adding $NH_3$ to the solution

(D) Adding $Mg(NO_3)_2$ to the solution

**41.** $\mathrm{Ba(IO_3)_2}(s) \ \rightleftarrows\ \mathrm{Ba^{2+}}(aq)\ +\ 2\ \mathrm{IO_3}^-(aq) \quad K_{sp} = 4 \times 10^{-9}$

According to the information about the dissolution of $\mathrm{Ba(IO_3)_2}(s)$ shown above, the correct value of S, the molar solubility of $\mathrm{Ba(IO_3)_2}(s)$, can be calculated using with of the following mathematical relationships?

(A) $\frac{1}{2}S = 4 \times 10^{-9} M$

(B) $2S = 4 \times 10^{-9}\ M$

(C) $S^2 = 4 \times 10^{-9}\ M$

(D) $4S^3 = 4 \times 10^{-9}\ M$ ✓

**42.** $\mathrm{Ag_2CO_3}(s) \rightleftarrows 2\ \mathrm{Ag^+}(aq) + \mathrm{CO_3}^{2-}(aq)$

The chemical equation above represents the equilibrium that exists in a saturated solution of $\mathrm{Ag_2CO_3}$. If $S$ represents the molar solubility of $\mathrm{Ag_2CO_3}$, which of the following mathematical expressions shows how to calculate $S$ based on $K_{sp}$ ?

(A) $S = \sqrt{K_{sp}}$

(B) $S = \sqrt{\frac{K_{sp}}{2}}$

(C) $S = \sqrt[3]{\frac{K_{sp}}{2}}$

(D) $S = \sqrt[3]{\frac{K_{sp}}{4}}$ ✓

## Chapter 10 Part 3: Ksp

**43.**

| **Solution** pH | $[Ag^+]_{eq}$ |
|---|---|
| 7.00 | $1.41 \times 10^{-4}$ |
| 8.00 | $1.41 \times 10^{-4}$ |
| 9.00 | $1.37 \times 10^{-4}$ |
| 10.00 | $1.00 \times 10^{-4}$ |
| 11.00 | $1.96 \times 10^{-5}$ |
| 12.00 | $2.00 \times 10^{-6}$ |
| 13.00 | $2.00 \times 10^{-7}$ |

A student investigates the effects of pH on the solubility of $AgOH(s)$, which dissolves in water according to the equation $AgOH(s) \rightleftarrows Ag^+(aq) + OH^-(aq)$. The value for $K_{sp}$ for $AgOH$ is $2.0 \times 10^{-8}$ at $298$ K. The student places the same mass of $AgOH(s)$ into $50.0$ mL of different solutions with specific pH values and measures the concentration of $Ag^+$ ions in each solution after equilibrium is reached. Based on the data in the table, what can be concluded about the solubility of $AgOH$?

(A) The solubility of $AgOH$ is unaffected by pH because $K_{sp}$ is a constant value at $298$ K.

(B) The solubility of $AgOH$ is unaffected by pH because it is the same value for pH 7.00 and pH 8.00.

(C) $AgOH$ is more soluble at higher pH because lower concentrations of $OH^-(aq)$ shift the solubility equilibrium toward $Ag^+(aq)$ and $OH^-(aq)$.

(D) $AgOH$ is less soluble at higher pH because higher concentrations of $OH^-(aq)$ shift the solubility equilibrium toward solid $AgOH$.

**44.**

$$CaF_2(s) \rightleftarrows Ca^{2+}(aq) + 2\,F^-(aq) \qquad K_{sp} = 3.9 \times 10^{-11}$$

$$HF(aq) \rightleftarrows H^+(aq) + F^-(aq) \qquad K_c = 6.8 \times 10^{-4}$$

The dissolution of calcium fluoride is represented by the equilibrium system above at $25^\circ C$. The $F^-$ ion is produced when the weak acid $HF$ dissociates. If solid calcium fluoride is added to equal volumes of the following solutions at $25^\circ C$, in which solution will the most calcium fluoride dissolve?

(A) Pure distilled water

(B) $1\ M\ HNO_3(aq)$

(C) $1\ M\ NaOH(aq)$

(D) A saturated aqueous $CaF_2$ solution

## Chapter 10 Part 3: Ksp

**45.** For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

A student is interested in studying the $Cu^{2+}(aq)$ ion, which is important in the environment.

$Cu(NO_3)_2(aq) + K_3PO_4(aq) \rightarrow$

(a) First, the student carries out a precipitation reaction. A solution of copper(II) nitrate is mixed with a solution of potassium phosphate, as represented above, and a blue precipitate forms. Write the balanced net ionic equation for the process.

(b) In a separate experiment the student decides to precipitate the $Cu^{2+}(aq)$ as $Cu(OH)_2(s)$. To make sure that $Cu(OH)_2(s)$ will precipitate, the student looks up the $K_{sp}$ value for $Cu(OH)_2(s)$, as shown below.

$Cu(OH)_2(s) \rightleftarrows Cu^{2+}(aq) + 2\ OH^-(aq)$ $\qquad K_{sp} = 2.2 \times 10^{-20}$

(i) Write the $K_{sp}$ expression for the dissolving of $Cu(OH)_2(s)$.

(ii) Calculate the concentration of $Cu^{2+}(aq)$ in a saturated $Cu(OH)_2$ solution in $mol/L$.

(c) The student incorrectly proposes that lowering the $pH$ of the saturated $Cu(OH)_2$ solution will increase the amount of precipitate that forms. Explain why the student is incorrect.

**Part A**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

✓

| 0 | 1 |
|---|---|

The response provides a correct balanced equation (state symbols not required).

- $3\ Cu^{2+}(aq) + 2\ PO_4{}^{3-}(aq) \rightarrow Cu_3(PO_4)_2(s)$

**Part B(i)**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

## Chapter 10 Part 3: Ksp

| 0 | 1 |
|---|---|

The response provides the correct expression.

- $K_{sp} = [Cu^{2+}][OH^-]^2$

**Part B(ii)**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides the correct calculated value consistent with part (b)(i).

- $K_{sp} = [Cu^{2+}][OH^-]^2$

  $2.2 \times 10^{-20} = (x)(2x)^2$

  $[Cu^{2+}] = x = 1.8 \times 10^{-7}\ M$

  OR

- $K_{sp} = [Cu^{2+}][OH^-]^2$

  $2.2 \times 10^{-20} = (x)(1.0 \times 10^{-7} + 2x)^2$

  $[Cu^{2+}] = x = 1.4 \times 10^{-7}\ M$

**Part C**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides a correct explanation.

- The amount of precipitate will decrease, not increase. When the $pH$ is lowered, the excess $H^+$ will react with $OH^-$, reducing $[OH^-]$ and therefore allowing additional $Cu^{2+}$ and $OH^-$ ions to enter solution, which increases the solubility of $Cu(OH)_2$.

# Chapter 10 Part 3: Ksp

**46.**

| Compound | $K_{sp}$ at 298 K |
|---|---|
| $Ag_2SO_4$ | $1 \times 10^{-5}$ |
| $PbSO_4$ | $1 \times 10^{-8}$ |

A 1.0 L solution $AgNO_3(aq)$ of and $Pb(NO_3)_2(aq)$ has a $Ag^+$ concentration of 0.020 $M$ and a $Pb^{2+}$ concentration of 0.0010 $M$. A 0.0010 mol sample of $K_2SO_4(s)$ is added to the solution. Based on the information in the table above, which of the following will occur? (Assume that the volume change of the solution is negligible.)

(A) No precipitate will form.

(B) Only $Ag_2SO_4(s)$ will precipitate.

(C) Only $PbSO_4(s)$ will precipitate. ✓

(D) Both $Ag_2SO_4(s)$ and $PbSO_4(s)$ will precipitate.

**47.**

| Compound | $K_{sp}$ |
|---|---|
| $Fe(OH)_2$ | $4.9 \times 10^{-17}$ |
| $Cu(OH)_2$ | $1.6 \times 10^{-19}$ |
| $Co(OH)_2$ | $1.1 \times 10^{-15}$ |

Which of the following ranks the compounds listed in the table above in order of increasing solubility?

(A) $Fe(OH)_2 < Cu(OH)_2 < Co(OH)_2$

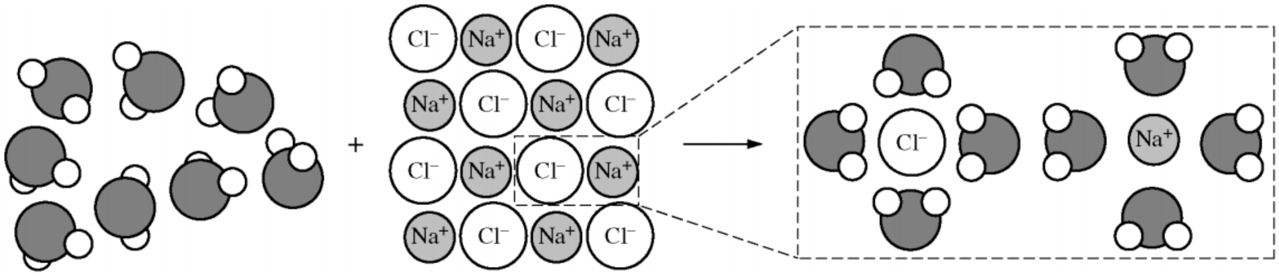
(B) $Fe(OH)_2 < Co(OH)_2 < Cu(OH)_2$

(C) $Co(OH)_2 < Fe(OH)_2 < Cu(OH)_2$

(D) $Cu(OH)_2 < Fe(OH)_2 < Co(OH)_2$ ✓

## Chapter 10 Part 3: Ksp

**48.**




The process of dissolution of NaCl($s$) in $H_2O(l)$ is represented in the diagram above. Which of the following summarizes the signs of ΔH° and ΔS° for each part of the dissolution process?

| | Breaking solvent-solvent interactions | | Breaking solute-solute interactions | | Forming solute-solvent interactions | |
|---|---|---|---|---|---|---|
| | $\Delta H°$ | $\Delta S°$ | $\Delta H°$ | $\Delta S°$ | $\Delta H°$ | $\Delta S°$ |
| (A) | + | + | + | + | – | – |
| (B) | + | + | + | + | – | + |
| (C) | – | – | – | – | + | + |
| (D) | – | + | – | + | + | – |

**49.**

| Solution | Solute | $K_{sp}$ at 25°C |
|---|---|---|
| X | AgBr | $5.0 \times 10^{-13}$ |
| Y | AgCl | $1.8 \times 10^{-10}$ |
| Z | AgI | $8.3 \times 10^{-17}$ |

Three saturated solutions (X, Y, and Z) are prepared at 25°C. Based on the information in the table above, which of the following lists the solutions in order of increasing [Ag+] ?

(A) X < Z < Y

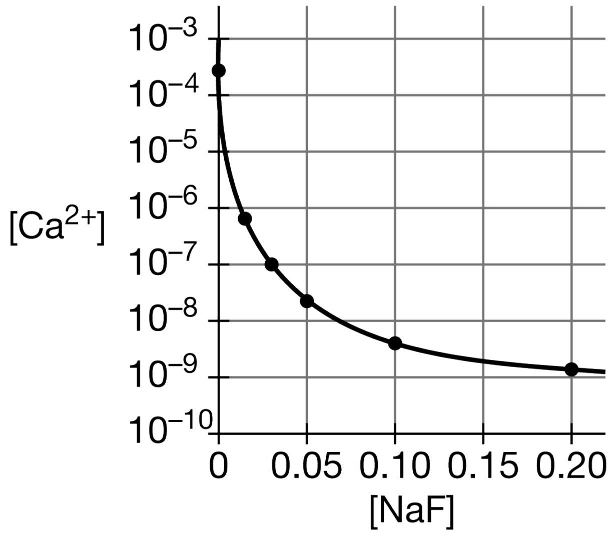
(B) Y < X < Z

(C) Z < Y < X

(D) Z < X < Y

## Chapter 10 Part 3: Ksp

**50.**




Which of the following statements best explains the trend in the molar solubility of $CaF_2(s)$ in various concentrations of $NaF(aq)$ shown in the graph above?

(A) Higher concentrations of $F^-(aq)$ allow less $CaF_2(s)$ to dissolve before the solution becomes saturated. ✓

(B) Higher concentrations of $Na^+(aq)$ lead to more favorable attractions between $Na^+(aq)$ ions and $F^-(aq)$ ions, thus less $CaF_2(s)$ dissolves.

(C) Higher concentrations of $F^-(aq)$ produce more $OH^-(aq)$ through hydrolysis, causing $Ca(OH)_2(s)$ to precipitate, thus more $CaF_2(s)$ dissolves.

(D) Higher concentrations of $F^-(aq)$ cause the formation of the ${CaF_4}^{2-}(aq)$ complex ion, which allows more $CaF_2(s)$ to dissolve.

**51.** $CuBr(s) \rightleftarrows Cu^+(aq) + Br^-(aq)$

Shown above is the chemical equation for the dissolution of the slightly soluble salt $CuBr(s)$. Its $K_{sp}$ value in pure water was experimentally determined. $CuBr$ was found to be much less soluble in a $0.001\ M$ NaBr solution than in pure water. Which of the following correctly explains the decrease in solubility of $CuBr$ in $0.001\ M$ NaBr?

(A) The concentration of water is much less in the $NaBr$ solution than in pure water, reducing the rate of dissolution of the salt.

(B) The presence of $Na^+$ ions in the solution inhibits the dissolution of salts.

(C) The presence of additional $Br^-$ already in the solution decreases the value of the $K_{sp}$ for $CuBr$, causing the solubility to decrease.

(D) The presence of additional $Br^-$ ions already in the solution means equilibrium will be reached when much less $CuBr$ has dissolved. ✓

## Chapter 10 Part 3: Ksp

**52.**

| Moles of $HCl$ added | Molar solubility of $MgCO_3$ |
|---|---|
| 0 | $2.6 \times 10^{-3}$ |
| $2.7 \times 10^{-4}$ | $3.3 \times 10^{-3}$ |
| $4.6 \times 10^{-4}$ | $5.5 \times 10^{-3}$ |
| $6.0 \times 10^{-4}$ | $6.9 \times 10^{-3}$ |

$$MgCO_3(s) \rightleftarrows Mg^{2+}(aq) + CO_3{}^{2-}(aq)$$

A saturated solution of $MgCO_3$ at equilibrium is represented by the equation above. Four different saturated solutions were prepared and kept at the same temperature. A given amount of $HCl$ was added to each solution and data were collected to calculate the molar solubility of $MgCO_3$ as shown in the table above. Which of the following can be concluded from the data?

(A) The molar solubility of $MgCO_3$ decreases with increasing acidity (lower $pH$).

(B) The molar solubility of $MgCO_3$ increases with increasing acidity (lower $pH$). ✓

(C) The $pH$ or acidity of the solution has a negligible effect on the molar solubility of $MgCO_3$.

(D) From the data it is not possible to determine the effect of changes in acidity on the molar solubility of $MgCO_3$.

## Chapter 10 Part 3: Ksp

**53.** **For parts of the free response question that require calculations, clearly show the method and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.**

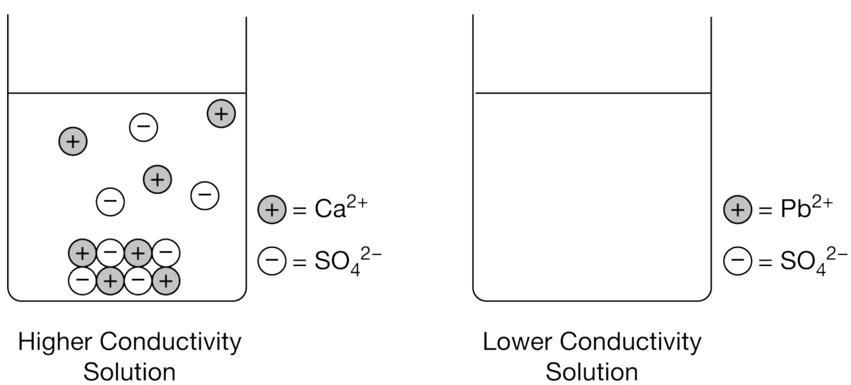
A student is studying the properties of $CaSO_4$ and $PbSO_4$. The student has samples of both compounds, which are white powders.

(a) The student tests the electrical conductivity of each solid and observes that neither solid conducts electricity. Describe the structures of the solids that account for their inability to conduct electricity.

The student places excess $CaSO_4(s)$ in a beaker containing $100\ mL$ of water and places excess $PbSO_4(s)$ in another beaker containing $100\ mL$ of water. The student stirs the contents of the beakers and then measures the electrical conductivity of the solution in each beaker. The student observes that the conductivity of the solution in the beaker containing the $CaSO_4(s)$is higher than the conductivity of the solution in the beaker containing the $PbSO_4(s)$.

(b) Which compound is more soluble in water, $CaSO_4(s)$ or $PbSO_4(s)$ ? Justify your answer based on the results of the conductivity test.

The left side of the diagram below shows a particulate representation of the contents of the beaker containing the $CaSO_4(s)$ from the solution conductivity experiment.




(c) Draw a particulate representation of $PbSO_4(s)$ and the ions dissolved in the solution in the beaker on the right in the diagram. Draw the particles to look like those shown to the right of the beaker. Draw an appropriate number of dissolved ions relative to the number of dissolved ions in the beaker on the left.

(d) The student attempts to increase the solubility of $CaSO_4(s)$ by adding $10.0\ mL$ of $2\ M\ H_2SO_4(aq)$ to the beaker, and observes that additional precipitate forms in the beaker. Explain this observation.

**Part A**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

## Chapter 10 Part 3: Ksp

| 0 | 1 |
|---|---|

The response provides a correct description.

· Ionic solids do not have free-moving ions that are required to carry an electric current. Therefore, there is no conduction of electricity.

**Part B**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides a correct answer and a valid justification.

· $CaSO_4$. The greater the electrical conductivity of the $CaSO_4$ solution relative to the $PbSO_4$ solution implies a higher concentration of ions, which comes from the dissolution (dissociation) of $CaSO_4$ to a greater extent.

**Part C**

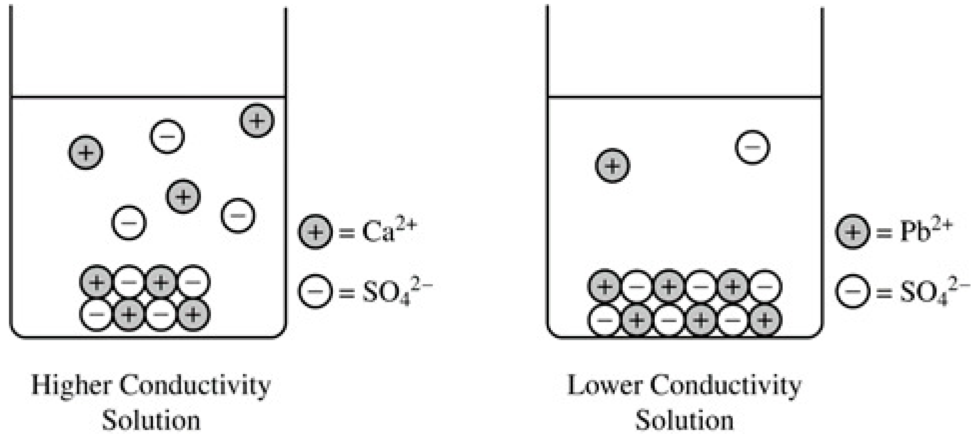
Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides a correct drawing that shows an equal number of cations and anions.

· The drawing shows solid $PbSO_4$ at the bottom of the beaker (similar to the solid shown for $CaSO_4$) and fewer dissociated $Pb^{2+}$ and $SO_4{}^{2-}$ ions in the solution.

## Chapter 10 Part 3: Ksp




### Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 |
|---|---|

The response provides a correct explanation.

· The additional precipitate is $CaSO_4$ that forms in response to the increased $[SO_4{}^{2-}]$ in solution. According to Le Chatelier's principle $(Q > K_{sp})$, the introduction of $SO_4{}^{2-}$ as a common ion shifts the equilibrium towards the formation of more $CaSO_4(s)$.